

The Market's Off-Frontier Component: Predictable but Not Priced

Timothy T. Simin

The Pennsylvania State University*

September 21, 2026

Abstract

Last month's market return forecasts this month's small-stock returns (Lo and MacKinlay, 1990). The forecast behaves as a near-arbitrage that frictions protect, not as a risk premium. Projecting the market return on a candidate pricing kernel splits the forecast into the component the kernel prices and the residual it discards. The residual contains all the predictability, for every kernel I try, including kernels built from test assets sharing no sorting variable with the residual. Conditioning the size frontier on the unpriced direction earns a real-time out-of-sample Sharpe ratio of 0.61, against -0.10 for the priced direction, yet the gain lives in microcaps and dies net of trading costs. Tests of conditional efficiency should ask where an improvement resides, not only whether it exists.

Keywords: conditional efficiency, stochastic discount factor, return predictability, lead-lag, limits to arbitrage.

JEL: G12, G14.

* An Internet Appendix with supplementary tables and details accompanies the paper. Data are from Kenneth French's data library, Amit Goyal's updated Goyal and Welch (2008) predictor series, He, Kelly, and Manela's intermediary capital factors, Pástor and Stambaugh's liquidity series, and J. Wurgler's sentiment data. I used a large language model (Claude, Anthropic) extensively as a research and editorial assistant in preparing this paper, including writing and debugging analysis code, executing data work under my direction, checking citations, and drafting and revising text. The research questions, design choices, interpretations, and conclusions are my own, and all model output was subject to my review. I take full responsibility for the content and for all remaining errors.

1 Introduction

Last month’s market return forecasts this month’s returns on small stocks. This lead–lag relation is thirty-five years old (Lo and MacKinlay, 1990), and a long debate has contested whether it is due to autocorrelation and nonsynchronous trading (Boudoukh et al., 1994), volume-paced adjustment (Brennan et al., 1993; Chordia and Swaminathan, 2000), or price delay from frictions (Hou and Moskowitz, 2005). What the debate never settled is what kind of predictability the lead–lag represents. Return predictability can be three things. It can be compensation for risk, priced because it lies in the span of the stochastic discount factor (SDF), durable because earning it means holding what others avoid, and bounded in Sharpe ratio by the price of that risk. It can be mispricing that arbitrage removes, unpriced but temporary, with out-of-sample failure and post-publication decay as its signatures (McLean and Pontiff, 2016; Martin and Nagel, 2022; Cho, 2020). Or it can be near-arbitrage protected by frictions, the paper’s case. The predictability is unpriced and yet persistent, with a gross Sharpe ratio too large to be a risk premium (Haddad et al., 2020), because microcap trading costs form a barrier exactly high enough to protect it. The friction does the work that risk does in the first case and that capital eventually does in the second. The raw forecast does not say which case obtains. For all the fact tells us, small stocks could simply be lagging their exposure to priced risk.

The three cases separate along two observable margins, where the forecastable variation lies relative to a pricing kernel and what survives out of sample and net of costs. Priced predictability is the first case. Unpriced predictability whose gain survives realistic costs is the second, and unpriced predictability whose gain dies at those costs is the third. This paper measures both margins for the lead–lag. In the Hansen and Richard (1987) framework, any return decomposes as $R = R^* + w R^{e*} + n$, where R^* and R^{e*} generate the mean–variance frontier, here the frontier of the five Fama–French factors, and n is a mean-zero residual orthogonal to the frontier. For the market *excess* return, a zero-cost payoff, the R^* term drops out, leaving $\text{Mkt} - \text{RF} = w R^{e*} + \eta$, with η the market’s own residual n . The decomposition separates a frontier component $w R^{e*}$ that a risk model prices from an off-frontier residual η that the model discards as noise. The forecasting power sits in η , while the priced frontier component forecasts nothing. None of fourteen canonical predictors forecasts the size cross-section. Only the off-frontier component does, outside the reach of the kernels that price that cross-section (Figure 1). Small stocks lag the part of aggregate returns those kernels do not price.

Not everything the paper says has the same standing. The priced share, the share of η ’s variance a kernel spans, is a statistical estimate, and “unpriced” is relative to the kernel doing

the pricing. The residual η is largely unpriced by the cross-sectional kernels a risk model would use, but it is 0.88-correlated with the market itself, and its priced share runs from 0% against the factor frontier to 77% against the CAPM, tracking each kernel’s market exposure. Only the first of those numbers is forced by the construction, since the factor frontier is the single direction η is built orthogonal to. The alphas earned by trading on η are evidence of a weaker sort. Timing a persistent signal guarantees part of any unconditional alpha, so I read the alphas as a cross-kernel pattern rather than as levels. Whether the alphas survive real-time implementation and trading costs is the last question, and that margin, not a premium, decides the classification (Section 6.2). From the standpoint of a forecaster η is the lagged market, no better and no worse (Section 4.2), and I do not claim it is a new predictor. The forecast is concentrated in the smallest decile, fades by the middle of the cross-section, and weakens after decimalization (Section 8.2). The contribution is neither a predictor nor a model-free pricing claim but the location of the predictability, and the diagnostic that delivers it.

The location matters because of what it implies for the frontier. It is not obvious that conditioning information can reside in a discarded residual. Filipović and Schneider (2025) prove that in a static cross-section an unpriced residual is equivalent to the factors spanning the SDF, so the residual cannot improve the mean–variance frontier. That equivalence, however, is a statement about the *unconditional* frontier. Conditionally, the classic point of Hansen and Richard (1987) and Ferson and Siegel (2001), an unconditional frontier is inefficient whenever some variable forecasts returns. If that variable is the residual itself, the information that would improve the frontier is exactly the part the model sets aside. The data show exactly that case, and the orthogonality is not a construction artifact. Against *disjoint* kernels, kernels built from assets that share no sorting variable with η , the residual is spanned no more than its market correlation mechanically implies. Strip the market from both η and each kernel, and the two stripped series do not co-move. This partial correlation is at most zero for every such kernel, and the raw priced share falls from 8.6% to 3.3% as added assets pull the kernel away from the market.

A block bootstrap turns the spanning statement into an interval. With 95% confidence no kernel spans more than 2.2% of η ’s market-neutral variation. Against the size–value kernel, a shrinkage-disciplined tangency portfolio of the 25 size–book-to-market portfolios and the paper’s headline SDF, 97.4% of η ’s variance is unspanned (Section 5.1). Splitting η into the part that the kernel spans and the unpriced remainder, conditioning the frontier of the size deciles on the unpriced part earns an out-of-sample Sharpe ratio of 0.61, while the priced part delivers -0.10 . A paired bootstrap places the gain where the location says it should be. Timing the unpriced component beats timing the priced one by 0.70 in annual

Sharpe (95% CI [0.34, 0.97]) and beats an implementable real-time static frontier by 0.45 ([0.11, 0.83]), both with $p \leq 0.005$. The asymmetry is conditional efficiency made concrete, with the value of conditioning sitting in the direction the pricing kernel ignores.

In gross terms the timing strategy earns a 19.1% annual alpha against the market and the size-value kernel jointly ($t = 4.1$), a magnitude Haddad et al. (2020) would call a near-arbitrage. The near-arbitrage label applies only to the gross object. The gain is hard to monetize, because it lives in the smallest, least liquid stocks, the breakeven one-way cost is about 123 basis points, and at realistic microcap spreads the net Sharpe ratio is zero or negative (Section 6.2). Frictions, not risk, bound the gain. Further tests identify the effect as the short-horizon, size-paced lead-lag of Lo and MacKinlay (1990) and Hou (2007), small stocks adjusting to last month’s zero-premium market move with a lag. Continuation at one month reverses by two years, indicating underreaction rather than overreaction (Hong and Stein, 1999). The tests do not resolve whether the lag is informational or liquidity-paced, but the classification does not rely on that distinction. The evidence places the effect in the third case, a near-arbitrage that frictions protect (Section 7.1).

The location is what changes the inference, and I quantify the consequences in Section 7.2. Without the decomposition, standard tools leave the lead-lag unclassified. A factor-timing rule on the lagged market succeeds, but a scaled-factor conditional model is rejected, and neither says which of the three cases holds. With the decomposition, the risk-based reading is bounded. If the lead-lag were a time-varying premium, the predictive slope would come through conditional beta times the conditional premium. In the data that channel generates at most a sixth of the observed slope with the market as the priced factor, and 1.5% of it with the size-value kernel. The rejected conditional model points the same way once its instrument, the lagged market, is split. The scaled factor built from the priced part of the instrument absorbs none of the alpha. All the absorption runs through the unpriced part, which is not the channel a risk story is allowed to use.

The diagnostic is not specific to the lead-lag. The split is an elementary orthogonal projection, but applied as a classifier it sorts predictability into the three cases. To show the diagnostic discriminates rather than labeling everything unpriced, I run it on predictors with opposite priors (Section 8.1). The diagnostic places the term spread’s predictability (Fama and French, 1989) in the priced span and finds the intermediary capital factor of He et al. (2017) several times more spanned by the size-value kernel than η . Both placements go in the direction theory expects. The implication is for how conditional efficiency is tested. Showing that conditioning information improves the frontier is not enough. Improvement in the priced span and improvement in the orthogonal span are observationally identical at the level of a Sharpe ratio but economically opposite, the first a risk-premium story and the

second, when it survives out of sample and dies net of costs, as here, a frictions story.

The paper sits at the junction of four literatures. Hansen and Richard (1987) and Ferson and Siegel (2001) characterize the conditional frontier and the optimal use of conditioning information, but do not ask where in the return space the value of conditioning resides. Kozak et al. (2018, 2020) and Haddad et al. (2020) bound predictability by no-arbitrage and find factor timing implies near-arbitrage Sharpe ratios, but work in the priced directions rather than locating predictability relative to the kernel. Favero et al. (2026) come closest, finding that past pricing errors, the deviations of an anomaly’s price from its model-implied value, forecast that anomaly’s own risk-adjusted return. Their mechanism is an error correction in price levels, asset by asset, whereas η is a within-period return component orthogonal to the frontier whose forecast is of *other* assets. Kim et al. (2021) apply a complementary conservatism in the cross-section, granting factor exposure its maximal explanatory weight before attributing alpha. The lead–lag literature the paper opened with remains active. Chinco et al. (2019) recover sparse lagged-return signals in the modern cross-section, Parsons et al. (2020) document a geographic lead–lag concentrated in small, thinly traded stocks, and Dong et al. (2022) run the mirror-image direction, the cross-section forecasting the market out of sample. None of that work locates the effect relative to a pricing kernel or connects it to frontier efficiency.

Daniel et al. (2020) hedge the unpriced components of factor returns to sharpen premia, Chernov et al. (2026) argue that unpriced common risks are pervasive enough to warrant rethinking cross-sectional asset pricing, and a contemporaneous literature splits returns into compensated and uncompensated parts, a transient month-end liquidity-demand concession (Nathan and Suominen, 2026) or systematic but unpriced taste risk (Saxena and Singh, 2026), characteristic by characteristic. The Hansen–Richard frontier supplies the geometric counterpart to those splits and ties the uncompensated part to conditional efficiency. (η is unpriced within a correct equity SDF, distinct from the benchmark misspecification of Greenwald, 2025.) Extending the diagnostic from the canonical time-series predictors studied here to the cross-sectional factor zoo (Feng et al., 2020; Bryzgalova et al., 2023; Jensen et al., 2023) is the next step. It would turn the location of predictability into a specification test that sorts the anomalies into overfitting (Chordia et al., 2020), priced risk (Daniel and Titman, 1997; Kelly et al., 2019), and friction-bound near-arbitrage (Chen and Velikov, 2023).

Section 2 constructs the decomposition and Section 3 describes the data. Section 4 shows that the off-frontier component carries the predictability and the frontier component none. Section 5 runs the priced/unpriced test (the first case versus the other two). Section 6 runs the out-of-sample and net-of-cost tests (the second case versus the third). Section 7

examines the mechanism and quantifies what the classification changes. Section 8 checks that the diagnostic discriminates and collects the robustness checks. Section 9 concludes.

2 The off-frontier component and its decomposition

2.1 The off-frontier component

Following Hansen and Richard (1987) and Cochrane (2005, ch. 5), any gross return decomposes as

$$R_i = R^* + w_i R^{e*} + n_i, \quad (1)$$

where R^* is the minimum-second-moment return, R^{e*} the excess return that represents means, w_i a scalar, and n_i an excess return with $E[n_i] = 0$ orthogonal to both R^* and R^{e*} . The coefficient on R^* is the payoff's price, so for a zero-cost *excess* return the R^* term drops out. The market excess return is such a payoff, and I use its two-term form directly.

In the standard L^2 geometry with inner product $E[xy]$, a unique payoff x^* in the span of traded payoffs represents prices, meaning the price of any payoff z is $p(z) = E[x^*z]$, and R^* is x^* divided by its own price $E[x^{*2}]$, which makes it a return. An excess return is a payoff with price zero, so $E[R^*z] = 0$ for every excess return z , and R^* is orthogonal to the entire excess-return space. Prices cannot tell excess returns apart, but means can. The payoff R^{e*} is the excess return that represents means in the same sense that x^* represents prices, $E[R^{e*}z] = E[z]$ for every excess return z . Projecting Mkt–RF on the two directions therefore puts zero weight on R^* and leaves the frontier part of the market's decomposition entirely to R^{e*} . The remainder is η , zero in mean and orthogonal to the frontier, the unpriced residual the paper studies.

Let R^e be the $T \times K$ matrix of the $K = 5$ Fama–French factor excess returns (Mkt–RF, SMB, HML, RMW, CMA) and $\mathbf{1}$ a T -vector of ones. The mean-representing excess return is the in-sample projection of the constant onto the factor span,

$$R^{e*} = R^e (R^{e\top} R^e)^{-1} R^{e\top} \mathbf{1}, \quad (2)$$

where $\top$ denotes the transpose, so that R^{e*} is the fitted value of a regression of $\mathbf{1}$ on the factors with no constant. With Mkt–RF the asset of interest, the *off-frontier component* is

$$\eta_t = (\text{Mkt} - \text{RF})_t - w R_t^{e*}, \quad w = \frac{\overline{\text{Mkt} - \text{RF}}}{\overline{R^{e*}}}, \quad (3)$$

The weight w is the ratio of the two sample means, so $\bar{\eta} = 0$ by construction. The same

choice makes η orthogonal to R^{e*} . Because R^{e*} represents means, $\overline{\eta R^{e*}} = \overline{\text{Mkt} - \text{RF}} - w \overline{R^{e*}}$, and the choice of w sets that difference to zero. Orthogonality to R^{e*} is orthogonality to the unconditional frontier, whose excess returns are the multiples of R^{e*} . Equation (3) is the two-term specialization of (1). Appendix A gives the full R^*/R^{e*} construction and the equivalence. A clarification on language, η is the part a risk model treats as “idiosyncratic,” in the sense of the Hansen–Richard frontier (orthogonal to the priced frontier R^{e*} , hence earning no premium), *not* statistically idiosyncratic. In fact η is 0.88-correlated with the market, since it is the market’s own off-frontier residual. It is unpriced, not uncorrelated.

2.2 Priced and unpriced components

The label “priced” is frequently applied to three distinct propositions. The first proposition is *spanning* of the state variable, meaning how much of the variation of η itself, the off-frontier return whose forecastable variation the paper locates, projects onto a candidate SDF return. This is the split $\eta = \eta_p + \eta_u$ constructed next. The second is *pricing of the managed payoff*, i.e. the object an investor would actually trade, the η -timed strategy $\eta_t R_{i,t+1}^e$ studied in Sections 6.1–6.2. I ask whether it earns an alpha against the kernel’s tangency return. A state variable can look spanned while the payoff that trades on it is not, and keeping the two apart resolves an apparent paradox later (Section 5.1). The third is an identified *cross-sectional price of η risk*, a λ_η estimated from mean returns. The paper’s evidence speaks to the three unequally, and the spanning statement is its centerpiece. Spanning is a second-moment quantity, identified in sample whether or not the kernel prices the cross-section well. The traded payoff alphas corroborate this. The risk-price reading is only weakly identified on these test assets (Section 8.3), and no conclusion rests on it.

To split η I project it on a stochastic discount factor, the tangency portfolio of the 25 size–book-to-market portfolios, estimated with the ridge shrinkage of Kozak et al. (2020). The RP-PCA of Lettau and Pelger (2020) is an alternative discipline, and the ridge is chosen for keeping the kernel a transparent function of μ and Σ . With μ and Σ the sample mean and covariance of the 25 excess returns and $\bar{d} = \text{tr}(\Sigma)/25$ their average eigenvalue,

$$b(\kappa) = (\Sigma + \kappa \bar{d} I)^{-1} \mu, \quad (4)$$

and the SDF return is $\text{MVE} = R_{25} b(\kappa)$, with R_{25} the $T \times 25$ matrix of the portfolio excess returns, scaled to unit variance. I fix the dimensionless κ ex ante at the conventional 0.01. A split-half validation check selects the same value (Appendix B), and the size–value priced/unpriced split is essentially invariant across the grid $\kappa \in \{0, 0.01, \dots, 20\}$ (priced share 2.6–3.0%), so nothing below depends on this choice. Note that κ does not enter the

construction of η or the size-decile forecasts of Section 4.1. It affects only the priced/unpriced split and the conditional timing. Regressing η on MVE with an intercept gives slope β and

$$\eta = \underbrace{\beta \text{ MVE}}_{\text{priced, } \eta_p} + \underbrace{(\eta - \beta \text{ MVE})}_{\text{unpriced, } \eta_u}. \quad (5)$$

The slope β comes from a regression with an intercept, so η_u has zero *covariance* with (zero beta on) MVE. Because $E[\eta] = 0$ while MVE has a nonzero mean, this is covariance-orthogonality, not orthogonality in the uncentered second-moment sense. The displayed components drop the intercept, so η_p and η_u carry equal and opposite means (the priced premium and its offset) while η_u bears no MVE risk. The split is relative to the candidate SDF, and Section 5.3 examines how it varies across kernels.¹

2.3 The location of the conditioning value

The split in (5) delivers a decomposition of the value of conditioning. Conditioning has first-order value only if η forecasts. The unconditional frontier of the size deciles is conditionally inefficient given η , in the mean-timing sense used throughout, if and only if some decile's conditional mean moves with η , $g_i \equiv \text{Cov}(\eta_t, R_{i,t+1}^e) \neq 0$ (Hansen and Richard, 1987; Ferson and Siegel, 2001), and the payoff that realizes the gain, $\eta_t R_{i,t+1}^e$, has unconditional mean exactly g_i because $E[\eta] = 0$. Since η_p and η_u are uncorrelated by construction, the gain splits additively. Slopes, predictive R^2 , and the covariances all decompose component by component, $g_i = g_{i,p} + g_{i,u}$ with $g_{i,k} = \text{Cov}(\eta_{k,t}, R_{i,t+1}^e)$, and the two pieces carry opposite economics. Variation indexed by η_p is a time-varying loading on the priced factor, a premium the kernel prices, while variation indexed by η_u improves the conditional frontier through a direction the kernel does not price. If $g_{i,p} = 0$ for every decile while some $g_i \neq 0$, the entire first-order value of conditioning is unpriced. Appendix A states this as Proposition 2 and gives the short proof, elementary orthogonal-decomposition algebra recorded to fix what the split does and does not establish, not as a theoretical contribution.

The construction fixes only the orthogonality, and only in the covariance metric ($E[\eta_p]$ and $E[\eta_u]$ are equal and opposite, not zero). It does not fix which component forecasts. In principle $g_{i,p}$ could absorb all of g_i , making the predictability a harvested risk premium.

¹Throughout, the “priced share” is $\text{var}(\eta_p)/\text{var}(\eta)$, the R^2 of this regression, which equals $\text{corr}(\eta, \text{MVE})^2$. The correlation enters squared because the share is a variance share. The unpriced part η_u is uncorrelated with MVE, so $\text{var}(\eta) = \text{var}(\eta_p) + \text{var}(\eta_u)$ and the priced and unpriced shares, corr^2 and $1 - \text{corr}^2$, sum to one. Unsquared correlations do not decompose additively this way. Squaring also removes the sign of the loading, which is irrelevant to spanning. Since $E[\eta] = 0$ by construction, the kernel has no premium on η to explain. The priced share instead measures how much of η 's *variation* lies in the direction the kernel compensates.

Which component carries the forecast is an empirical question. The predictive covariances are identified by the recursive forecasts of Sections 4.1–4.2, the split by sample second moments relative to a stated kernel (Section 2.2), and the one channel the algebra leaves open, whether the timed payoff loads on the kernel through conditional second moments, is measured directly from the traded payoff’s alphas in Section 5.1 rather than asserted. The data give the polar answer, $g_{i,p} \approx 0$ and $g_i \approx g_{i,u}$ (Table 3, Panel B), so the conditioning information the unconditional frontier discards lies entirely in the direction the kernel ignores. The magnitudes are economic rather than marginal. The smallest decile moves 1.31% per month per standard deviation of the lagged standardized market, η ’s forecasting equivalent (Table 7), and the timing gain is worth 1.7 to 5.8% per year in certainty-equivalent terms gross of costs and essentially zero net (Sections 6.1–6.2).

2.4 Recursive estimation and inference

For every out-of-sample statistic, I re-estimate the above through date t only. Starting from an initial window of $t_0 = 240$ months and expanding, I recompute at each t R^{e*} , w , and η from (2)–(3) on $R_{1:t}^e$, the SDF weights $b(\kappa)$ from (4) on $R_{25,1:t}$, and β from (5). The period- t values then enter the time- $t+1$ forecast, so no period- t predictor uses later data. Predictive regressions use Newey–West standard errors with 12 lags, and out-of-sample forecasts are compared to the prevailing (expanding-window) mean using the out-of-sample R^2 and Clark–West statistic of Appendix B. The recursion makes the results out of sample in the coefficient sense. The factor *definitions* are not, since the choice of which characteristics form the factors was made with knowledge of the history, and Section 5.3 bounds what that remaining look-ahead can contribute.

2.5 Testing hierarchy

Three primary claims each rest on one designated statistic. The location claim rests on the smallest decile’s Clark–West statistic (3.9, Table 1), the unpricedness claim on the market-neutral partial correlation against the largest disjoint kernel ($\rho = -0.14$, bootstrap $p(\rho > 0) = 0.027$, Table 6), and the conditioning claim on the paired Sharpe increment of the unpriced over the priced direction (0.70, bootstrap $p = 0.001$, Table 9). With one-sided normal p -values for Clark–West, a Holm correction² across the three families leaves all three

²The Holm correction controls the familywise error rate, the probability of at least one false rejection in a family of m tests, and is valid under arbitrary dependence among the tests (Holm, 1979). Order the p -values from smallest to largest and compare the j th to $\alpha/(m - j + 1)$, rejecting until the first failure. The adjusted p -value reported in the text is the j th ordered p -value multiplied by $m - j + 1$, with monotonicity across j enforced, so a claim survives at level α when its adjusted p is below α . The procedure is uniformly

significant (adjusted p below 0.001, 0.002, and 0.027), and within the decile family the smallest decile survives a Holm correction across all ten (adjusted p below 0.001), decile two survives at 0.03, and no larger decile does, which is the gradient the mechanism predicts. The paper’s other statistics belong to none of these families and are supporting evidence. The industry, sentiment, horizon, and dispersion results (Section 7.1, Internet Appendix) corroborate the mechanism. The fourteen-predictor battery (Section 8.1) and the placebo draws (Section 5.2) are negative controls, where running more tests gives more chances for a spurious winner and none appears. The kernel and span families support universal statements, all nine disjoint kernels leave ρ at or below zero, so adding kernels makes those claims harder to sustain rather than easier, and the familywise bound is the 2.2% ceiling of Table 6. The shrinkage grid, the block-length sweep, and the subperiod cuts re-estimate the same statistics rather than test new hypotheses.

3 Data

The sample is monthly, July 1963–April 2026 (754 months), set by the start of the five-factor data. Series are from Kenneth French’s data library unless noted. Portfolio returns are value-weighted and converted to excess returns by subtracting the one-month Treasury rate (the RF series in the factor file). Values coded -99.99 or -999 are treated as missing. All inputs are public. A replication package with the data construction and code accompanies the paper.

The factors are the five Fama–French factors and RF and portfolio data are from Ken French’s web page. I use both the monthly and the daily counterparts. The five factor excess returns are the asset space R^e in (2). The test portfolios are the value-weighted 25 size–book-to-market portfolios are the SDF base assets in (4). The ten size deciles, the ten beta deciles, and the 49 industries portfolios are value-weighted, both monthly and daily. Industry average firm size, used in Section 7.1, is the “Average Firm Size” block of the industry file. Sentiment is the macro-orthogonalized Baker and Wurgler (2006) index (SENT_ORTH) from J. Wurgler’s data page, monthly 1965:07–2023:12. For the *realized second moments* of the Internet Appendix I form within-month realized quantities from daily data (realized market variance, off-frontier volatility (*dis*), aggregate idiosyncratic volatility, and cross-sectional dispersion, with formulas there). Months with fewer than 15 daily observations or fewer than 40 industries reporting are dropped, setting the realized sample to 1968:06–2026:04. The fourteen aggregate predictors of Section 8.1 are the Goyal and Welch (2008) set, from Amit Goyal’s website, updated through 2025:12 (Goyal et al., 2024). The intermediary

more powerful than the Bonferroni correction, which multiplies every p -value by m .

capital risk factor is from Zhiguo He’s website. The traded liquidity factor is Pástor and Stambaugh’s posted series.

4 The off-frontier component carries the predictability

4.1 The off-frontier component predicts the size cross-section

The off-frontier component forecasts size-decile returns out of sample, concentrated entirely in small stocks. Table 1 reports, for each decile, the out-of-sample R^2 and Clark–West statistic from a recursive predictive regression of next-month excess returns on η_t , with R^{e*} , w , and the predictive coefficient re-estimated through date t , alongside the in-sample slope t -statistic. The in-sample slope’s t -statistic declines from 5.5 in the smallest decile to 1.2 in the largest, and the out-of-sample R^2 falls from +3.7% to −0.3%. The largest decile (essentially the value-weighted market) shows nothing. Because I examine ten deciles, 49 industries, and several sentiment regimes, some individual statistics will be large by chance. The primary inference therefore rests on the designated statistics and familywise corrections of the testing hierarchy in Section 2.5, and the monotone size gradient and consistency across cuts corroborate it. The η loading is perfectly rank-ordered by size (Spearman −1.00), the small-minus-big loading carries a Newey–West t of 5.25, and the rank correlation stays below −0.6 in every stationary block-bootstrap resample (design in Section 5.2).

Two cuts that do not sort on size mechanically show where the predictability comes from. Sorting on market beta yields no predictability in any decile (out-of-sample R^2 near zero throughout, no Clark–West above 1.6). Sorting the 49 industries by firm size instead reproduces the gradient. In tercile-portfolio regressions the in-sample slope on η falls from $t = 2.8$ in small-firm industries to 0.9 in large-firm ones (Internet Appendix Table IA.4 reports the underlying industry-level loadings), and only the small-firm-industry portfolio forecasts out of sample (Clark–West 1.7, against 0.3 for large-firm industries; Section 7.1). The industry out-of-sample R^2 stays far below the smallest decile’s, because even small-firm industries average \$180–330M per firm (an order of magnitude above microcaps), so the effect concentrates in the very smallest stocks. The source is the size-paced lead–lag effect of Lo and MacKinlay (1990) (Section 7.1), not exposure to systematic risk.

4.2 The frontier component forecasts nothing

Decomposing the market excess return into its priced frontier component $w R^{e*}$ and its unpriced off-frontier component η yields the affirmative finding the raw lead–lag does not. *The market forecasts the size cross-section through its unpriced component, not its priced*

one. Table 2 sets the two pieces against each other. Its η columns reproduce the gradient of Table 1, and the new fact is in the frontier-component columns (labeled $\text{mkt} \perp \eta$, the part of the market orthogonal to η , which equals $w R^{e*}$). In the joint in-sample regression of each decile on both lagged pieces, the frontier component is insignificant in every decile (no t above 1.7 in absolute value) while η 's slope keeps its gradient. In separate single-predictor recursive forecasts, the frontier component's out-of-sample R^2 is negative for every small decile (-2.7% at decile 1) and zero at the largest. Economically, small stocks lag the part of aggregate returns that carries no premium. They underreact to the idiosyncratic-labeled component of market moves, not to lagged exposure to priced risk. The cross-section's predictability by the market sits, at its source, outside the priced span.

This also explains why η , taken as a standalone forecast, is no better than the lagged market. Because η is 0.88-correlated with the contemporaneous market, using η_t to predict month $t+1$ is, mechanically, using the lagged market. In a recursive horse race for decile 1, lagged Mkt-RF alone gives an OOS R^2 of 2.9% and η alone 3.7%, but η adds nothing to the lagged market (Clark-West -0.2), and in the joint regression the two are symmetric substitutes (slopes 0.16, $t = 1.6$ and 0.17, $t = 1.7$). The forecast itself is not new. What is new is its location. The lagged market predicts small stocks only through its off-frontier component.

Is that a mechanical consequence of the Hansen-Richard construction, so that the off-frontier residual of *any* factor would forecast the cross-section? No (Internet Appendix Table IA.1). Forming $\eta_i = f_i - w_i R^{e*}$ for each of the five factors, with f_i the factor's excess return and w_i its coefficient on R^{e*} , the market's off-frontier component is the dominant positive predictor of small-stock returns (decile-1 OOS R^2 3.65%, in-sample $t = 5.5$, with the size gradient). The off-frontier part of SMB (itself a size factor) adds an independent positive piece (1.35%, $t = 4.3$). Those of HML, RMW, and CMA forecast with the *wrong* (negative) sign, their strength tracking only their negative correlation with the market direction. The predictability is specific to the *aggregate* (and size) off-frontier component, exactly what a lead-lag in which small stocks slowly absorb market-wide news implies (Section 7.1), and not a byproduct of the decomposition. Nor does a simpler residual do the same work. The market orthogonalized to the other four factors by regression forecasts nothing (decile-1 out-of-sample R^2 -0.6%), so the Hansen-Richard construction retains the market content that carries the lead-lag, which a regression residual discards.

5 Is the predictability priced?

5.1 The predictability is unpriced by cross-sectional kernels

The forecastable variation lies in the SDF-orthogonal direction. Against the headline size–value SDF, 97.4% of η is orthogonal to the kernel (Table 3, Panel A). That number is less informative than it looks. The size–value tangency is itself near-market-neutral, and a kernel with little market exposure leaves most of any market-like return unspanned, whatever that return’s construction. The stronger evidence comes from a kernel that shares nothing with η ’s construction. A tangency SDF built from industry-, beta-, and momentum-sorted portfolios (no size or book-to-market) prices only 8.6% of η ’s variation. Section 5.2 shows that neither share is forced by the construction. The conclusion is the same under either kernel. The predictability is in the orthogonal direction. Because $E[\eta] = 0$ the two components carry equal and opposite mean returns (+3.0% and –3.0% per year), and splitting the predictor (Panel B), the unpriced component reproduces the full first-decile out-of-sample R^2 (3.8%, Clark–West 4.0) while the priced component has a negative out-of-sample R^2 in every decile. This is the polar case of Proposition 2 in Appendix A, with $g_{i,p} \approx 0$ and $g_i \approx g_{i,u}$, so the conditioning value sits entirely in the SDF-orthogonal direction.

The split and the alpha tests price different objects. The split in equation (5) is a statement about the *state variable* η , namely how much of its variation is spanned by the SDF return. The object an investor ultimately cares about is the *managed payoff*, i.e. the conditional return $\eta_t R_{i,t+1}^e$ of the η -timed strategy, and whether it earns an alpha the kernel cannot explain. The two questions are linked only under two auxiliary conditions. The payoff’s conditional covariance with the kernel must not move with the signal, and the signal must not forecast the kernel (the conditions noted alongside Proposition 2 in Appendix A). Neither holds automatically, so the payoff-space question needs its own test.

Table 4 runs it directly. The traded payoff $z_{t-1} R_{1,t}^e$, with z the standardized η (the η -timed smallest decile, with mean 15.4% per year and Sharpe 0.64)³ earns an annualized alpha of 16–20% against *every* candidate kernel’s tangency return, with t between 5.0 and 6.2 (e.g. 17.3% against the size–value SDF, 19.2% against the disjoint kernel). The 19.1% alpha the introduction and Section 6.2 lead with is this same payoff standardized in real time and priced against the market and the SDF *jointly* rather than the SDF alone. It falls inside this 16–20% band, at a lower t of 4.1 because the real-time object is noisier.

³This managed payoff standardizes η over the full sample, as do the alphas in Table 4. Its gross Sharpe ratio is 0.64. The real-time, out-of-sample traded strategy of Section 6.2 (Table 10) standardizes η recursively and has a gross Sharpe ratio of 0.63. The two numbers describe the same strategy under in-sample versus real-time standardization, not a discrepancy.

These are alphas against *unconditional* kernels, and a managed payoff that times a persistent signal will mechanically earn a large unconditional alpha against any of them, so the levels are partly guaranteed and not the point. The informative content is the cross-kernel *pattern*. The alpha barely moves (16–20%) even as the predictor-space priced share ranges from 0 to 77%, because timing is a conditional, nonlinear operation. Weighting next month’s return by this month’s η produces a payoff with a market beta of -0.43 (-0.45 for the real-time version of Section 6.2), essentially market-neutral, so the market exposure that makes the *state variable* look priced is washed out of the *payoff*.⁴ The timed payoff is near-orthogonal to every kernel, and the unpricedness is not an artifact of one of them. I lead with the predictor-space split because it distinctly separates the priced and unpriced directions. The pattern of the payoff-space alphas corroborates it in the object investors trade.

5.2 What the construction forces and what it does not

How much of the unpricedness in Section 5.1 is forced by construction? Some of it is an identity. The residual η is a traded zero-cost return with $E[\eta] = 0$, and a traded factor’s price of risk is its mean, so $\lambda_\eta = 0$ by construction and “ η is unpriced” in the premium sense is not a finding. The paper’s weight rests on two facts that are not definitional. First, the predictability sits in η rather than in the priced frontier component $w R^{e*}$, when it could have been the reverse (Section 4.2). Second, the orthogonality survives kernels not used to construct η . The construction makes η orthogonal to one direction only, R^{e*} , so the priced share against R^{e*} is 0.0% by design (Table 5, last row) and the priced share against every other kernel is empirical. Empirical is not yet cross-sectional, however. Because η is 0.88-correlated with the market, any kernel with market exposure prices η mechanically whether or not it prices η ’s market-neutral variation, and shrinkage that tilts the tangency toward the market raises that exposure. Proposition 1 separates the two channels.

Proposition 1. *Let m be a candidate SDF return, R_m the market excess return, and decompose η and m into their market and market-orthogonal parts, $\eta = a R_m + \eta^\perp$ and $m = c R_m + m^\perp$ with $\eta^\perp, m^\perp \perp R_m$. With $r_{\eta M} = \text{corr}(\eta, R_m)$, $r_{mM} = \text{corr}(m, R_m)$, and the partial correlation $\rho = \text{corr}(\eta^\perp, m^\perp)$,*

$$\text{corr}(\eta, m) = r_{\eta M} r_{mM} + \rho \sqrt{(1 - r_{\eta M}^2)(1 - r_{mM}^2)}. \quad (6)$$

⁴This resolves an apparent paradox in Table 4. The CAPM rejects the payoff as strongly as any kernel (alpha 18.5%, and the largest t in the table, 6.2), yet it is the kernel against which the state variable looks most priced, since 77% of η ’s variance is market variance. The CAPM’s only priced direction is the market, and the timed payoff carries almost none.

The priced share $\text{corr}(\eta, m)^2$ then splits into a market-induced term $r_{\eta M}^2 r_{mM}^2$, its value when $\rho = 0$, and a cross-sectional term governed by ρ . The kernel spans η 's market-neutral variation, pricing η beyond its market exposure, if and only if $\rho > 0$. Provided $\text{corr}(\eta, m) \geq 0$, as holds for every kernel below, this is equivalent to the priced share exceeding the market-implied share $r_{\eta M}^2 r_{mM}^2$.

Proof. Demean and scale η , m , R_m to unit variance and regress the first two on R_m : $\hat{\eta} = r_{\eta M} \hat{R}_m + \sqrt{1 - r_{\eta M}^2} \hat{\eta}^\perp$ and $\hat{m} = r_{mM} \hat{R}_m + \sqrt{1 - r_{mM}^2} \hat{m}^\perp$, with unit-variance residuals $\hat{\eta}^\perp, \hat{m}^\perp \perp \hat{R}_m$. Taking the expectation of the product, the cross terms in $\hat{R}_m$ vanish and $E[\hat{\eta}^\perp \hat{m}^\perp] = \rho$, which is (6). $\square$

The proposition redirects the test. The mechanical benchmark is the market-implied share $r_{\eta M}^2 r_{mM}^2$, not the overlap with R^{e*} , and the object that measures cross-sectional pricing is ρ . Shrinkage that tilts the tangency toward the market raises r_{mM}^2 and with it both shares, so the raw share cannot settle the question and ρ can. For every disjoint kernel, a tangency built from test assets sorted on none of η 's construction inputs, $\rho \leq 0$ (Table 6, last column). It is -0.09 for the 67-portfolio industry/beta/momentum kernel, -0.14 for the 97-portfolio kernel that adds short-term reversal, long-term reversal, and residual variance, and negative for every kernel between. Equivalently, the raw priced share lies *below* the market-implied share in every case (8.6% against 11.0% at 67 assets, 3.3% against 6.0% at 97). No kernel spans η 's market-neutral variation. Enlarging the kernel from 67 to 97 portfolios makes this visible. It lowers the tangency's market exposure (14.3% $\rightarrow$ 7.8%), the raw priced share falls with it (8.6% $\rightarrow$ 3.3%), and ρ stays near zero.

The sign of ρ is a more reliable statistic than the raw priced share against a single kernel, for two reasons. First, it is insensitive to the shrinkage κ that makes the raw priced share fragile. Raising κ from 0.01 to 1 lifts the raw share from 8.6% to 33% by tilting the tangency toward the market, but ρ stays negative throughout.⁵ Second, the two results, a non-positive ρ and a raw priced share that falls as the kernel enlarges, are estimated precisely enough to survive inference. A stationary block bootstrap (mean block six months, 5,000 replications, re-estimating η , the tangency, and the market residualization on each resample) puts the 97-portfolio partial correlation at -0.14 (95% CI $[-0.26, 0.00]$, $p(\rho > 0) = 0.03$) and the enlargement decline in the raw priced share at 5.3 points (95% CI $[0.8, 6.9]$, $p < 0.001$). The decline is significant and the market-neutral correlation is not positive. Table 6 reports the same interval for every kernel. All nine point estimates are negative, four intervals lie

⁵Across $\kappa \in \{0, 0.01, 0.2, 1\}$ the disjoint priced share is 7.8, 8.6, 17, 33%, tracking the tangency's rising market exposure, while the market-neutral ρ is $-0.08, -0.09, -0.12, -0.21$. The split-half validation check selects $\kappa = 0.01$ (Appendix B), but by Proposition 1 the cross-sectional conclusion, the sign of ρ , does not rely on that choice.

strictly below zero, and the upper end of the 95% interval for the largest of the nine partial correlations is 0.15, so even the most favorable kernel spans at most 2.2% of η 's market-neutral variation. The partial correlation is also robust to the orthogonalization anchor.⁶ By contrast the raw single-kernel share is imprecise (8.6%, 95% CI [1.3, 11.2]) and its excess over the kernel's 6.3% overlap with R^{e*} is not distinguishable from zero ($p = 0.51$), which is why that comparison is neither the right benchmark nor the evidence.

A placebo test confirms that the pattern is not an artifact of the zero-mean decomposition. I draw 1,000 random zero-mean returns in the five-factor span and push each through the identical fully recursive pipeline. Generic draws carry little forecast (median out-of-sample R^2 1.1%, against the 3.8% the same pipeline gives η), draws matched to η 's market correlation reach a median of 2.7% with η at the 73rd percentile, and their priced shares and partial correlations are indistinguishable from η 's. The split does not manufacture predictability. The predictability of η is its lead-lag market content, and its unpricedness is the expected behavior of the market's own residual.

5.3 Unpriced relative to which kernels?

The priced share is *kernel-dependent* (Table 5). It is 0.0% against the frontier R^{e*} , at most about 0.5% against a consumption-based CRRA SDF, 2.6% against size-value, 8.6% against the disjoint kernel, 15–18% against industries and momentum, 37% against beta deciles, and 77% against the market itself. The ordering follows each kernel's market exposure (the table's last column) almost monotonically, as Proposition 1 implies for the market's own residual. What “unpriced” means is therefore class-relative. The residual η is largely unpriced by the disciplined cross-sectional and consumption kernels a risk model would use to price the size cross-section, but *not* by the market itself (77%). The admissible class is set ex ante by the question, disciplined tangency kernels of cross-sectional test assets built as in equation (4). The priced share is a spanning quantity, identified in sample whether or not the kernel prices the cross-section correctly.

Not every row of Table 5 carries the same weight. The CAPM and beta-decile rows are diagnostics of the market-exposure channel, not members of the admissible class. The consumption row deserves the least weight. The same CRRA kernel prices the market itself near zero (the equity-premium puzzle), so it would assign almost any equity payoff a near-zero share, and I report it only as a disjoint-inputs check. The weight rests on the disjoint-kernel

⁶Its bootstrap standard error is 0.07, so the 97-asset $\rho = -0.14$ sits about two standard errors below zero, and controlling for SMB and HML in addition to the market leaves it negative (-0.11 and -0.14). Controlling for the five-factor tangency R^{e*} instead is vacuous, since η is orthogonal to R^{e*} by construction and residualizing on it returns the raw correlation.

family of Table 6 and the out-of-sample conditioning gain, not on any single kernel’s priced share. What survives every caveat is that the forecastable variation, and the conditioning gain it carries, sit in the direction these kernels leave essentially unpriced.

The decomposition (and hence the location claim) is defined relative to the FF5 excess-return span. To check that “off-frontier” is not an artifact of that geometry, I rebuild η from alternative spans and rerun the two central tests (Internet Appendix Table IA.2). Whether the frontier is FF3, FF5, FF6 (FF5 plus momentum), or FF5 augmented with the 49 industry portfolios, η remains essentially the same object (0.80 to 1.00 correlated with the FF5 η and 0.70 to 0.96 correlated with the market). It still forecasts the smallest size decile out of sample (OOS R^2 3.0–3.7%, Clark–West 3.4–3.9), and it stays largely unpriced against the size–value SDF (priced share 0–9.5%, rising mildly as the span absorbs more market exposure). The exact 0.0% for FF3 is not greater unpricedness but a near-mechanical orthogonality. The size–value tangency is closely aligned with the FF3 frontier’s own R^{e*} , to which η is orthogonal by construction. The informative entries are the nonzero priced shares for the richer spans, whose construction frontier no longer spans the size–value kernel. The location result does not depend on the FF5 frontier.

The FF3 row can be tightened further. The recursion of Section 2.4 makes every coefficient out of sample, but the factor *definitions* were fixed with knowledge of the history, in 1993 for the three-factor model and 2015 for the five (Fama and French, 1993, 2015). Rebuilding η from the FF3 frontier and evaluating only on 1994:01–2026:04, months entirely after those definitions were fixed, gives a smallest-decile out-of-sample R^2 of 2.7% (Clark–West 2.9), and on that window alone the residual remains unpriced, with a priced share below 1% and $\rho \leq -0.09$ against both the size–value and disjoint kernels.

6 Can the gain be harvested?

6.1 The value of conditioning lies in the SDF-orthogonal direction

Conditioning the size-decile frontier on the unpriced component improves it out of sample. Conditioning on the priced component does not. I form real-time conditional cross-sectional strategies that weight each decile by its predicted return, with η , the SDF, the priced/unpriced split, and the predictive loadings all estimated through date t . Table 9 reports the out-of-sample Sharpe ratio of each. These weights are the cross-sectionally demeaned predicted returns. They sum to zero across deciles and so define a transparent, zero-cost managed overlay, a self-financing tilt added on top of the deciles. The overlay is not the full Ferson and Siegel (2001) conditional tangency, which would require an estimated

conditional covariance matrix. A profitable zero-cost overlay is enough to establish that the unconditional frontier is *not* conditionally efficient, and the demeaned-prediction rule has the advantage that its priced and unpriced versions differ only in the conditioning direction, isolating the contribution I want to measure. Any profitable zero-cost addition to the set of assets proves the unconditional frontier is conditionally inefficient, i.e. if the frontier were efficient, no signal-driven tilt could improve it. So, the overlay’s Sharpe is a lower bound on the conditioning gain located by direction, rather than as the conditional tangency itself, which I report separately below.

The unpriced direction delivers the entire conditioning value, an out-of-sample annual Sharpe ratio of 0.61 (i.i.d. $t = 3.96$, Newey–West $t = 5.2$), and the priced direction delivers nothing (-0.10). What matters is the *increment* from conditioning rather than the level against zero, so I pair the realized monthly series and block-bootstrap the Sharpe difference (mean block six months, 5,000 replications, a common resample index across the paired series). Two paired increments are decisive. First, timing the unpriced component beats timing the priced component by 0.70 in annual Sharpe (95% CI $[0.34, 0.97]$, $p = 0.001$). The two strategies share an asset menu and an estimator and differ only in the conditioning direction, so the increment isolates the location claim from estimation method.

Second, the unpriced overlay beats an *implementable* real-time static frontier, the recursively estimated (κ -shrunk) size-decile tangency whose own out-of-sample Sharpe is 0.16, by 0.45 ($[0.11, 0.83]$, $p = 0.005$). The static frontier is barely implementable because $\Sigma^{-1}\mu$ on the near-collinear deciles is estimation-wrecked, whereas the overlay never inverts Σ . Against an *oracle* static frontier handed full-sample weights (out-of-sample Sharpe 0.56) the increment is 0.05 and not distinguishable from zero ($[-0.34, 0.47]$, $p = 0.38$), but that bar grants the static frontier the look-ahead the overlay is denied, and I do not rest the claim on it. Both increments are insensitive to block length. Across mean blocks of 3–36 months the location increment’s 95% interval ranges from $[0.34, 0.99]$ to $[0.38, 1.02]$ with $p \leq 0.001$ throughout, the increment over the implementable frontier from $[0.09, 0.86]$ to $[0.16, 0.80]$ with $p \leq 0.009$, and the level interval for the unpriced strategy ($[0.38, 0.91]$ at the baseline) from $[0.35, 0.90]$ to $[0.43, 0.93]$, all $p(\text{SR} \leq 0) < 0.001$. In sample the priced direction appears to contribute (55% of the expansion), but that is overfit and vanishes out of sample. The unconditional frontier is conditionally inefficient, and the information that improves it lies in the unpriced span.

The conditioning result invites the same scrutiny as the forecast (Section 4.2). Is the gain specific to η_u , or is it again the lagged market acting on small stocks? Conditioning on the lagged market directly (Table 9, first row) delivers the same gain (an out-of-sample Sharpe ratio of 0.70, if anything marginally above η_u ’s 0.61), and the two timing strategies

are 0.81 correlated. The higher Sharpe is not a higher mean. The lagged-market strategy earns essentially the same annual return as the η_u strategy (14.8% versus 15.0%) at slightly lower volatility, so the lagged market is at most marginally better, consistent with η_u being its active ingredient rather than a rival to it. Conditioning on the priced component returns a Sharpe ratio of -0.10 , and the priced-component strategy is only 0.32 correlated with the lagged-market strategy against 0.81 for the unpriced. The market’s conditioning value for the size frontier lives in its unpriced span. The priced span contributes nothing out of sample.

The conditional inefficiency is economically important but not a free lunch. Measured against the *implementable* static frontier (out-of-sample Sharpe 0.16), the value of conditioning is the 0.45 paired increment above ($p = 0.005$). Against the frontier’s non-implementable in-sample best case (0.53) it is higher but not separably so, so the conditional-efficiency claim rests on the implementable comparison and on the location of the gain. In certainty-equivalent terms the gross gain is worth 1.7 to 5.8% per year to a mean–variance investor (risk aversion 10 down to 3). Because the paper quotes Sharpe ratios of several distinct objects, Appendix Table 11 collects each with its construction, information set, and sample.

The volatility-timing literature has the same two halves, conditioning gains that are valuable on paper (Moreira and Muir, 2019) but fragile out of sample (Cederburg et al., 2020) and largely erased by costs and limits to arbitrage (Barroso and Detzel, 2021). Here the decomposition adds the location of the gain, which is what classifies it. The off-frontier residual carries about 77% of the market’s return variance (the squared market correlation, Internet Appendix), so the variation a risk model labels idiosyncratic is most of the market, and it carries the entire conditioning gain. The frontier’s only improvable direction is one that cross-sectional kernels do not price. Whether the gain survives costs is the question of Section 6.2.

The demeaned overlay is transparent but not the optimal use of the signal. As a check, the Ferson and Siegel (2001) optimal conditional weights raise the conditional Sharpe ratio to 0.70 and leave the location unchanged. Conditioning the optimal weights on the unpriced component alone delivers 0.67, on the priced component alone 0.30, and a paired block bootstrap puts the gap at 0.37 (95% interval $[0.03, 0.71]$, $p = 0.014$, details in the Internet Appendix). The gain is *larger* under optimal weights and the bulk of it still lies in the unpriced span, so the transparent overlay remains the primary object.

6.2 Near-arbitrage, bounded by frictions

In gross terms the conditioning gain is large. Net of costs it disappears. The out-of-sample η -timed small-decile strategy earns an alpha of 19.1% per year ($t = 4.1$) against the market and the disciplined SDF jointly, a magnitude Haddad et al. (2020) read as near-arbitrage, a label I use for the gross alpha and not for a tradeable return. Its loadings are 0.003 on the kernel and -0.45 on the market. The alpha is a kernel-adjusted intercept, distinct from the strategy’s raw performance. The payoff’s gross mean return is 16.8% per year (Sharpe 0.63), and the alpha sits above it because hedging out the market’s premium at a -0.45 beta increases the intercept. The gross Sharpe, the 123-basis-point breakeven, and the net-of-cost ladder below are properties of the raw return and do not move with the kernel.

The alpha and the conditioning gains of Section 6.1 are gross, and gross gains are exactly what net-of-cost model comparison (Detzel et al., 2023) and the implementable-frontier program of Jensen et al. (2026) discipline. This one does not survive it. The strategy turns over its position roughly once a month in the smallest, least liquid decile (Table 10). Its breakeven one-way cost is about 123 basis points, and its net Sharpe ratio is 0.37 at 50 basis points, 0.12 at 100, and negative at 200. All cost figures are *one-way* (half-spread) charges per unit of turnover. The short leg is largely untradeable (Novy-Marx and Velikov, 2016), and short-sale costs alone eliminate many anomaly alphas (Muravyev et al., 2025), but that constraint is not the binding one here. The strategy is short the smallest decile in 46% of months, an untradeable microcap short carrying 48% of the turnover for about a quarter of the gross gain, and dropping it *raises* the breakeven one-way cost from 123 to 172 basis points. The binding friction is the microcap trading cost on both legs.

The relevant benchmark for the breakeven is therefore the measured one-way effective cost of trading the smallest stocks, and the measured series bracket the breakeven from above for most of the sample. Gibbs estimates of the one-way effective cost for the smallest quartile run from roughly 100 to 200 basis points on the NYSE from the 1960s through the 1990s, higher still on Amex and Nasdaq (small-Nasdaq costs peak above 300 basis points in the early 1990s), and fall below 50 basis points only by the mid-2000s (Hasbrouck, 2009). The median round-trip effective spread at the microcap boundary (the 2,000th largest firm) falls from about 4% in the 1960s to about 1% in the 2000s, one-way roughly 200 down to 50 basis points (Novy-Marx and Velikov, 2016), and the average effective spread paid by equal-weighted, microcap-tilted academic implementations is 206 basis points in sample and 68 after 2005, one-way about 103 and 34 (Chen and Velikov, 2023). Against these series the 123-basis-point breakeven sits at or below the one-way microcap cost for most of 1963–2000 and above it after decimalization, the same era pattern as the gross effect itself, and Section 8.2 nets the strategy era by era. Two caveats attach to the post-decimalization

comparison. Low-frequency spread proxies overstate TAQ effective costs after 2003 by 25–50 basis points, and none of these measures include the price impact of trading a value-weighted microcap decile (Chen and Velikov, 2023; Patton and Weller, 2020).

The gain lives where the friction does, in the microcap decile. The out-of-sample forecast is overwhelmingly there (an R^2 of 3.65% for the smallest decile against 0.19% for an equal-weighted portfolio of deciles two through ten), and the conditional-timing Sharpe ratio falls from 0.59 to 0.43 when the smallest decile is dropped, surviving only because deciles two and three retain a weaker lead–lag. That residual gain is also friction-bound. The ex-microcap overlay turns over so much that its breakeven one-way cost is only 22 basis points, so even at a 25-basis-point cost, well below small-stock spreads for most of the sample, its net Sharpe ratio is negative. The microcap concentration places the effect in the family Hou et al. (2020) document, anomalies that live in the smallest stocks and fade under cost scrutiny (Novy-Marx and Velikov, 2016; DeMiguel et al., 2020; Novy-Marx and Velikov, 2022), the signature of a friction-bound effect and the reverse of a risk premium. Kang et al. (2026) find that when the cross-section is conditioned on a *priced* state (the option-implied risk premium of Martin, 2017), the conditional relation *strengthens* once microcaps are excluded, because risk compensation does not live in the least liquid stocks. The gain from η does the opposite.

The near-arbitrage is what frictions, not risk, leave on the table, and the reading is deliberately two-sided. The strategy is not worth trading, since net of costs its Sharpe ratio is indistinguishable from zero, but the large gross gain says what the predictability is, unpriced variation that only frictions keep from being competed away. This classification rests on two things I test (the predictability is unpriced and its net-of-cost Sharpe is zero), not on post-publication survival, which I do not measure. A pre- versus post-Lo and MacKinlay (1990) comparison is confounded because the effect was documented early in the sample, so I leave that distinction to the taxonomy of Internet Appendix Table IA.7 rather than claiming it as evidence.

7 Mechanism and interpretation

7.1 Mechanism: size-concentrated lead–lag, not sentiment or risk

The predictability has the signature of a gradual, size-paced lead–lag, small and illiquid assets adjusting to aggregate moves with a lag (Lo and MacKinlay, 1990; Hou, 2007), not of sentiment-driven mispricing (Baker and Wurgler, 2006) or risk. Six tests support that reading, and I begin with the one that most directly rules out the closest competing explanation,

the month-end liquidity demand of Nathan and Suominen (2026).

That literature traces much of the factor zoo to a transient, predictable price concession in the days before month-end, when investors raising cash sell the holdings cheapest to liquidate, a friction-bound, uncompensated, small-stock effect that could be confused with η 's. It is not the same effect. Split each month's smallest-decile excess return into its last six trading days (about 29% of the month) and the rest. In that split η predicts *none* of the month-end portion (slope -0.004 , $t = -0.15$, -1% of the full-month slope) and *all* of the rest (slope 0.33 , $t = 6.8$, out-of-sample R^2 5.3% against 3.7% for the full month on the same daily-built series, with η re-estimated recursively). The predictability is spread across the following month, a gradual adjustment, not a month-end cash demand that reverses within days.

A lagged adjustment implies underreaction, a response that continues at short horizons and reverses once the move is absorbed, whereas overreaction reverses from the start. The response of the smallest decile to a one-standard-deviation η shock is $+130$ basis points at one month ($t = 5.5$), fades to zero by month four, and turns negative at months nine and eighteen. Cumulatively it builds to $+1.7\%$ by month three and reverses to -1.5% by month twenty-four ($t = -1.6$). Figure 2 plots the path. Short-run continuation giving way to long-run reversal rules out overreaction. It does not by itself separate slow information diffusion from liquidity-paced adjustment.

Adjustment at different speeds makes a further prediction that mean-timing alone does not. If assets absorb an aggregate η move at different speeds, the cross-section should spread out in the month after a shock, so η 's realized volatility, $dis = sd(\eta)$, should forecast the next period's cross-sectional dispersion of industry returns. It does, out of sample (Internet Appendix Table IA.6). The measure dis adds to a forecast of industry-return dispersion beyond an autoregression and aggregate idiosyncratic volatility (Clark–West 4.1). The increment is modest and dis is collinear with aggregate idiosyncratic volatility, so I read it as corroboration rather than a second result, a distinct moment of the same adjustment that follows from diffusion but not obviously from a risk-premium or sentiment account.

Across the 49 industries (a cut that does not sort on size mechanically) the *in-sample* predictive loading on η falls monotonically across firm-size terciles, from 0.0073 in small-firm industries to 0.0022 in large-firm ones, and the cross-industry correlation between loading and log firm size is -0.61 (Internet Appendix Tables IA.4 and IA.5, all computed on the same 1963:07–2026:04 sample). Firm size is each industry's average per-firm market equity (Kenneth French's *Average Firm Size*), averaged over the sample in logs. It spans more than an order of magnitude, from about $\$180$ – $\$330$ M per firm in the smallest-firm industries (real estate, fabricated products, rubber, textiles, toys) to $\$3$ – $\$6.5$ B in the largest (tobacco,

telecom, beer), and a paired Newey–West test on the monthly log gap separates any two industries reliably (the typical drug firm is 5.6 times the typical toy firm, $t = 23$). The industries that lag η most are fabricated products, real estate, textiles, entertainment, and health, and those that lag least are drugs, food, oil, tobacco, and gold. The gradient is an in-sample pattern. Out of sample the industry predictability survives only weakly and only for the smallest-firm industries (Section 4.1), because even small-firm industries are an order of magnitude larger than microcaps. The lead–lag is present across industries (Hong et al., 2000; Hou, 2007), not an artifact of size-sorted portfolios.

If sentiment drove the effect, it would concentrate in high-sentiment periods. It does the opposite. In a predictive regression of the smallest decile’s return on η , the macro-orthogonalized Baker–Wurgler index, and their interaction (Internet Appendix Table IA.3), the sentiment level enters with the sign Baker and Wurgler (2006) document (high sentiment forecasts lower small-stock returns, -0.0071 , $t = -2.7$) but the interaction is negative (-0.0048 , $t = -2.3$). The slope on η is 0.020 ($t = 4.5$) when sentiment is below its median and 0.009 ($t = 4.0$) when above, and η and sentiment are nearly orthogonal (correlation -0.12). Sentiment moves small-stock returns through its own channel but neither subsumes nor amplifies the lead–lag.

Nor is the predictability a repackaging of aggregate liquidity. Adding the Pástor and Stambaugh (2003) liquidity level to the smallest-decile forecast leaves η ’s slope essentially unchanged (t from 5.6 to 5.1 on the common sample), the liquidity level does not forecast on its own in the joint regression ($t = -0.3$), and the interaction is insignificant ($t = 1.1$), so the lead–lag is not simply stronger when aggregate liquidity is low (η is only 0.25-correlated with the liquidity level). The friction that bounds the gain is the *cross-sectional* cost of trading microcaps (Section 6.2), not the aggregate liquidity state.

These tests converge on a size-concentrated, frictional lead–lag. An effect that reverses at long horizons, is widest in small-firm industries, and is if anything stronger when sentiment is low describes aggregate moves reaching illiquid assets slowly, and read with Section 5.1, small assets lag the *unpriced* component of those moves. Firm size proxies both the information environment and arbitrage costs, so the tests separate the lead–lag from sentiment and from risk but not slow information diffusion (Hong and Stein, 1999; Hong et al., 2000) from a pure liquidity channel. That ambiguity is immaterial to the paper’s claim. Under either reading the gain is a friction-bound, unpriced near-arbitrage, not risk and not a still-open mispricing.

7.2 An interpretation horse race

What would a careful researcher conclude about the lead-lag from standard tools, using the lagged market alone and no decomposition? Factor timing would come first, and it works. Timing the size deciles on the lagged market earns an out-of-sample Sharpe ratio of 0.70 (Table 9), and nothing in that number says whether the gain is a harvested conditional premium or a friction-bound near-arbitrage. The two standard next steps are a conditional-beta calculation and a scaled-factor pricing test, and Table 7 runs both, instrumented by the lagged standardized market z^m as that researcher would.

Panel A bounds the risk-based reading. If the lead-lag were time-varying priced risk, the predictive slope would come through $\beta_t \lambda_t$ with $\beta_t = b_0 + b_1 z_t^m$ and $\lambda_t = \mu_0 + \mu_1 z_t^m$, so the implied slope is $b_1 \mu_0 + b_0 \mu_1$. The observed slope is 1.31% per month per standard deviation of the lagged market ($t = 5.5$). With the market as the priced factor, both risk ingredients are statistically zero ($t = 1.4$ for b_1 , 0.9 for μ_1) and their point estimates combine to 0.22% per month, a sixth of the observed slope, most of it (0.19 of the 0.22) the market's own weak autocorrelation passed through a static beta rather than beta variation. With the disciplined kernel as the priced factor the channel delivers 1.5% of the slope. The conditional-risk reading is therefore bounded at a sixth of the effect under the most generous point estimates and at a sixtieth under the kernel a researcher would use to price this cross-section. Setting both insignificant ingredients at their one-sided 95% upper bounds at once raises the implied share to 46%, so even in the case least favorable to my reading, risk accounts for less than half of the slope.

Panel B runs the conventional conditional-model test. Does adding the scaled factor $z_t^m \cdot f_{t+1}$ price the lagged-market-managed payoff $z_t^m R_{1,t+1}^e$? The conditional model is rejected, and more strongly than the unconditional one (the alpha falls from 19.1% to 13.4% per year but its t rises from 5.3 to 7.6). What the rejection cannot do is locate the anomaly. A rejected conditional model is routinely read as a call for a richer one, an escape always available in principle because conditional models are untestable without specifying the information set (Hansen and Richard, 1987). The decomposition closes that escape by splitting the instrument into its priced-state and unpriced content. The scaled factor built from the priced state absorbs nothing (the alpha is unchanged, -3% absorbed), all the absorption the conditional model achieves runs through the unpriced content (31%), and against the disciplined kernel no scaled factor absorbs more than about a tenth. The only channel a time-varying-premium story allows, conditioning information about the priced state, is the channel that does no work. Without the split the researcher ends with an unlocated anomaly and an open invitation to keep searching for the conditional model that prices it. With it, a priced-state scaled factor can absorb the payoff only through conditional-second-moment

channels, and those channels are empty here.

8 Generality and robustness

8.1 A located-predictability battery

Every application of the split so far has returned the same verdict, “unpriced,” and a diagnostic that can return only one answer is uninformative. To establish that the split has content, I run it across the fourteen Goyal and Welch (2008) predictors of the aggregate market (updated through 2025; Goyal et al., 2024), asking of each the two questions asked of η . Does its predictability sit in the priced or the unpriced direction (Table 8, Panel A), and does it forecast the size cross-section, priced or unpriced (Panel B)? On the first question, a predictor of the time-varying *risk premium* should forecast the priced frontier component $w R^{e*}$, not the off-frontier η that carries the lead-lag (Section 4.2). The term spread, the canonical such predictor (Fama and French, 1989), does exactly that. It forecasts the priced component (in-sample $t = 2.75$, out-of-sample R^2 0.26%) and not the unpriced one ($t = 0.06$, -0.97%), the mirror image of η . The other business-cycle predictors (the default yield spread, the long-term return) load on the priced side in sample. The valuation ratios carry little monthly predictive power (Goyal and Welch, 2008) and discriminate weakly either way, a property of those signals rather than of the test. The diagnostic returns “priced” for predictors whose predictability should be priced and “unpriced” for η .

The stronger question is which of these predictors forecast the small-stock cross-section at all, and whether they do so priced. *None* of the fourteen forecasts the smallest size decile out of sample (the best, the default return spread, reaches 0.11%, most are negative, and no Clark–West statistic exceeds 1.5). Only the off-frontier component does (out-of-sample R^2 3.06% on the common sample), and its managed payoff earns an 18.2% annual alpha against the disciplined SDF ($t = 3.85$), in the unpriced direction. The lagged market, η ’s proxy, behaves identically (3.02%, alpha 20.2%, $t = 3.77$). Figure 1 plots Panel B. The fourteen predictors cluster at zero forecasting power, η and the lagged market sit alone in the forecasts-and-unpriced corner, and the forecasts-but-priced corner, where a risk-premium account would put the effect, is empty. Unpriced cross-sectional predictability is therefore not a generic property of aggregate predictors. This is what the split adds to “the lagged market forecasts small stocks.” It classifies the forecast as unpriced and shows the property is specific, neither of which the raw signal reveals.

A complementary leg uses a priced *factor* rather than a predictor and tests the diagnostic’s other facet, spanning. A factor priced through directions the kernel spans should carry

a high priced share, and the intermediary capital risk factor of He et al. (2017), which prices the size, value, and many other cross-sections, does. Its priced share against the size–value SDF is 11.6%, four times η ’s 2.8% over the same 1970–2025 sample.⁷ The gap is not market exposure, since the factor is *less* correlated with the market than η (0.75 versus 0.88), but loading on the cross-sectional, non-market SDF directions. Across the two legs, a priced *predictor*’s forecast lands in the priced direction and a priced *factor* is spanned by the kernel, while η does neither. The metric detects only pricing the kernel itself captures. Traded liquidity (Pástor and Stambaugh, 2003), a priced factor orthogonal to size and value, is *not* spanned by this SDF and so is not flagged, a scope limit of the kernel, not of the diagnostic. One column of the taxonomy remains unvalidated. The mispricing case, whose signature is post-publication decay, has no positive control here, since the canonical aggregate predictors offer no consensus example of a decayed monthly forecast. A decayed-anomaly control belongs to the cross-sectional factor zoo, where documented decay exists.

8.2 Robustness

The oldest critique of the lead–lag holds that small stocks are forecast by their *own* past return rather than by the market’s. On this view the lead–lag is largely the small portfolio’s own autocorrelation, produced by nonsynchronous trading and partial adjustment (Boudoukh et al., 1994). The critique has force here, and I do not claim to separate the two signals. The smallest decile’s own lagged return is 0.66-correlated with η . Used alone, it forecasts the decile as well as η does (out-of-sample R^2 of 3.61% against 3.65%). In a joint in-sample regression both are significant (η $t = 2.5$, own lag $t = 3.4$), but out of sample adding η to the own-lag forecast yields no significant improvement (Clark–West 1.4). The paper’s question is whether the predictability is priced, not which signal carries it, and on that question the own lag gives the same answer as η . The own lag’s priced share against the size–value kernel is 10.2% (2.6% for η), and its managed payoff earns a 17.8% annual alpha against the disciplined SDF (Newey–West $t = 5.0$), indistinguishable from η ’s 17.3%. Under either signal the predictability sits outside the priced span. The friction-bound classification is unchanged as well, since own autocorrelation in a value-weighted microcap portfolio is itself a friction phenomenon.

The off-frontier component’s forecast of the smallest decile is present across subperiods but not constant. It is stronger before decimalization (in-sample $t = 5.4$, out-of-sample R^2

⁷Both priced shares are $\text{corr}(\cdot, \text{MVE})^2$ computed on the common 1970:01–2025:05 sample set by the He et al. (2017) factor’s availability, with η and the size–value SDF re-estimated on that window. Against the same SDF over η ’s full sample (1963:07–2026:04) η ’s priced share is 2.6% (Table 5), so the 2.8% here is the common-sample figure, not the full-sample one.

4.5% pre-2001) and weaker after ($t = 2.5$, 2.3%), as a friction-bound reading predicts when microcap spreads fall. The attenuation is not specific to the lead-lag. Index-level serial dependence weakened and flipped sign around the world over the same period (Baltussen et al., 2019), and structural breaks in premia cluster in the early 2000s (Smith and Timmermann, 2022). The costing aligns era by era (all cost figures one-way). Pre-decimalization the timed strategy grosses an annual Sharpe ratio of 0.75 but nets -0.1 to -0.4 at that era’s measured microcap costs of 150–200 basis points (Hasbrouck, 2009; Novy-Marx and Velikov, 2016). Post-decimalization it grosses 0.53 and nets $+0.06$ at 100 basis points. A material net gain survives only under the most optimistic modern cost assumptions (0.3–0.4 at 25–50 basis points), which sit at the bottom of the measured post-2005 range of roughly 30–70 basis points (Section 6.2) and ignore price impact in a value-weighted microcap decile. The gross effect and the friction fell together, and at each era’s plausible costs the net gain stays near zero.

The predictability is not bid-ask bounce or a stale-price artifact of the month-turn. Skipping the first trading day of the forecast month barely moves the in-sample t ($4.6 \rightarrow 4.4$ on the daily-built decile series, whose full-month statistics sit slightly below the monthly series of Table 1), and skipping a full week leaves it significant ($t = 4.0$). A one-day reversal would not survive. Its economic magnitude is front-loaded, however. The out-of-sample R^2 falls from 3.3% to 0.5% once the first week is skipped, so most of the tradeable content is realized within days, a fast frictional adjustment early in the month. That is close to stale-price territory, and one more reason the right reading is friction-bound rather than tradeable mispricing.

8.3 What does not work

Two natural extensions fail, and reporting them clarifies what the paper does and does not claim. A cross-sectional price of η risk cannot be identified on standard test assets. A two-pass Fama–MacBeth regression on the 25 size-book-to-market and 49 industry portfolios, with η as a factor, returns a price of risk indistinguishable from zero and a cross-sectional R^2 of about 0.03. On these assets the model does not identify risk prices at all (the caution of Lewellen et al. (2010)). Of the three readings separated in Section 2.2, the *risk-price* reading (λ_η) is the one this cross-section cannot pin down, and the paper does not lean on it. The *spanning* reading asks how much of η ’s variance the kernel’s tangency return spans. That is a second moment, identified in sample whether or not the kernel prices the cross-section correctly, and the disjoint-kernel priced share of Section 5.1 is a statistic of that kind. The out-of-sample conditioning gain of Section 6.1, a managed-payoff result, likewise requires no

identified price of risk. Weak identification of λ_η bounds the risk-price interpretation, not the location result. The Internet Appendix reports the two-pass estimates, the price of market risk, the price of η risk, and the cross-sectional R^2 , with Shanken standard errors.

The off-frontier volatility does not forecast the level of the covariance matrix. Contemporaneously *dis* and the Frobenius norm of the rolling industry covariance move closely together (log-log correlation 0.87), but they are two measurements of the same aggregate volatility, not a forecast and its target. A predictive regression of the norm on lagged *dis* is significant only when both are built from the same backward-looking rolling window, and vanishes ($t \approx 0$) once the covariance is measured as a forward, non-overlapping realized quantity from daily returns. The one second-moment forecast that survives (Internet Appendix) is of cross-sectional *dispersion*, a forward object outside *dis*'s information set. More generally, a predictor and a target built from the same window will look linked whether or not a forecast relationship exists.

9 Conclusion

The lead-lag of Lo and MacKinlay (1990) is a near-arbitrage that frictions protect, not a risk premium and not a mispricing awaiting capital. The classification rests on two measured margins, where the forecastable variation lies relative to a pricing kernel and what survives net of costs. On the first margin, splitting the market return into the component a kernel spans and the residual it discards places all of the forecast in the residual, and the priced component forecasts nothing. Against the size-value kernel 97.4% of the residual's variance is unspanned, and with 95% confidence no kernel built from assets that share no sorting variable with the residual spans more than 2.2% of its market-neutral variation. Conditioning the size-decile frontier on the unpriced direction earns a real-time out-of-sample Sharpe ratio of 0.61, against -0.10 for the priced direction. The unconditional frontier is therefore conditionally inefficient, and the information that improves it lies where the kernel does not price. On the second margin, the gain lives in the smallest decile, its breakeven one-way cost is about 123 basis points, and at measured microcap costs the net Sharpe ratio is zero or negative. The gross effect and the friction fell together after decimalization, as a friction-bound reading predicts.

The finding is a location, not a new predictor. As a forecaster the residual is the lagged market, no better and no worse, and what the paper adds is where the forecast sits and a diagnostic that finds it. "Unpriced" is relative to the kernel doing the pricing. The residual's priced share runs from 0% against the factor frontier to 77% against the CAPM and tracks each kernel's market exposure, so the claim is that the kernels a risk model would use for the

cross-section do not price the residual, not that the residual is uncorrelated with the market. The mechanism is a gradual, size-paced lead-lag in which small, illiquid stocks underreact to the zero-premium part of aggregate moves. Sentiment neither subsumes nor amplifies it, and the evidence does not say whether the lag is informational or liquidity-driven, a distinction the classification does not rely on. The design establishes the location, the friction bound, and a ceiling on the risk-premium reading, which conditional-beta estimates put at a sixth of the lead-lag slope. It does not deliver a model-free resolution of the risk-versus-frictions debate, and the classification is where the measured margins put the weight.

Tests of conditional efficiency should ask where an improvement resides, not only whether it exists. Improvement in the priced span is a risk-premium story, while improvement in the orthogonal span that survives out of sample and dies net of costs, as here, is a frictions story. Standard conditional tests do not draw the distinction. Instrumented with the lagged market they reject the conditional model without saying why. The split draws it for any predictor or factor that registers at the monthly frequency and loads on directions the kernel can see, and its verdict is not automatic. Run across the fourteen Goyal and Welch (2008) predictors it places the term spread’s predictability in the priced span and finds none that forecasts the size cross-section, so “unpriced” is not a label the test attaches to everything. The next step is to take the diagnostic to the cross-sectional factor zoo. Locating each anomaly’s predictability relative to the kernel and asking what survives out of sample and net of costs would sort anomalies into overfitting, priced risk, and friction-bound near-arbitrage, a distinction the proliferation of machine-learned discount factors and return forecasts (Gu et al., 2020; Chen et al., 2024; Kelly et al., 2024) has made pressing.

References

- Baker, Malcolm, and Jeffrey Wurgler. 2006. “Investor Sentiment and the Cross-Section of Stock Returns.” *Journal of Finance* 61(4): 1645–1680.
- Baltussen, Guido, Sjoerd van Bakkum, and Zhi Da. 2019. “Indexing and Stock Market Serial Dependence Around the World.” *Journal of Financial Economics* 132(1): 26–48.
- Barroso, Pedro, and Andrew Detzel. 2021. “Do Limits to Arbitrage Explain the Benefits of Volatility-Managed Portfolios?” *Journal of Financial Economics* 140(3): 744–767.
- Boudoukh, Jacob, Matthew P. Richardson, and Robert F. Whitelaw. 1994. “A Tale of Three Schools: Insights on Autocorrelations of Short-Horizon Stock Returns.” *Review of Financial Studies* 7(3): 539–573.
- Brennan, Michael J., Narasimhan Jegadeesh, and Bhaskaran Swaminathan. 1993. “Investment Analysis and the Adjustment of Stock Prices to Common Information.” *Review of Financial Studies* 6(4): 799–824.
- Bryzgalova, Svetlana, Jiantao Huang, and Christian Julliard. 2023. “Bayesian Solutions for the Factor Zoo: We Just Ran Two Quadrillion Models.” *Journal of Finance* 78(1): 487–557.
- Cederburg, Scott, Michael S. O’Doherty, Feifei Wang, and Xuemin (Sterling) Yan. 2020. “On the Performance of Volatility-Managed Portfolios.” *Journal of Financial Economics* 138(1): 95–117.
- Chen, Andrew Y., and Mihail Velikov. 2023. “Zeroing In on the Expected Returns of Anomalies.” *Journal of Financial and Quantitative Analysis* 58(3): 968–1004.
- Chen, Luyang, Markus Pelger, and Jason Zhu. 2024. “Deep Learning in Asset Pricing.” *Management Science* 70(2): 714–750.
- Chernov, Mikhail, Magnus Dahlquist, and Lars A. Lochstoer. 2026. “Unpriced Common Risks: Rethinking Cross-Sectional Asset Pricing.” *Annual Review of Financial Economics*, forthcoming. NBER Working Paper 34009.
- Chinco, Alex, Adam D. Clark-Joseph, and Mao Ye. 2019. “Sparse Signals in the Cross-Section of Returns.” *Journal of Finance* 74(1): 449–492.
- Cho, Thummim. 2020. “Turning Alphas into Betas: Arbitrage and Endogenous Risk.” *Journal of Financial Economics* 137(2): 550–570.
- Chordia, Tarun, Amit Goyal, and Alessio Saretto. 2020. “Anomalies and False Rejections.” *Review of Financial Studies* 33(5): 2134–2179.

- Chordia, Tarun, and Bhaskaran Swaminathan. 2000. “Trading Volume and Cross-Autocorrelations in Stock Returns.” *Journal of Finance* 55(2): 913–935.
- Cochrane, John H. 2005. *Asset Pricing*. Revised ed. Princeton University Press.
- Daniel, Kent, Lira Mota, Simon Rottke, and Tano Santos. 2020. “The Cross-Section of Risk and Returns.” *Review of Financial Studies* 33(5): 1927–1979.
- Daniel, Kent, and Sheridan Titman. 1997. “Evidence on the Characteristics of Cross Sectional Variation in Stock Returns.” *Journal of Finance* 52(1): 1–33.
- DeMiguel, Victor, Alberto Martín-Utrera, Francisco J. Nogales, and Raman Uppal. 2020. “A Transaction-Cost Perspective on the Multitude of Firm Characteristics.” *Review of Financial Studies* 33(5): 2180–2222.
- Detzel, Andrew, Robert Novy-Marx, and Mihail Velikov. 2023. “Model Comparison with Transaction Costs.” *Journal of Finance* 78(3): 1743–1775.
- Dong, Xi, Yan Li, David E. Rapach, and Guofu Zhou. 2022. “Anomalies and the Expected Market Return.” *Journal of Finance* 77(1): 639–681.
- Fama, Eugene F., and Kenneth R. French. 1989. “Business Conditions and Expected Returns on Stocks and Bonds.” *Journal of Financial Economics* 25(1): 23–49.
- Fama, Eugene F., and Kenneth R. French. 1993. “Common Risk Factors in the Returns on Stocks and Bonds.” *Journal of Financial Economics* 33(1): 3–56.
- Fama, Eugene F., and Kenneth R. French. 2015. “A Five-Factor Asset Pricing Model.” *Journal of Financial Economics* 116(1): 1–22.
- Favero, Carlo A., Alessandro Melone, and Andrea Tamoni. 2026. “Reversal Patterns in Risk-Adjusted Anomaly Returns.” Working paper; SSRN Working Paper 4516438, last revised January 2026.
- Feng, Guanhao, Stefano Giglio, and Dacheng Xiu. 2020. “Taming the Factor Zoo: A Test of New Factors.” *Journal of Finance* 75(3): 1327–1370.
- Ferson, Wayne E., and Andrew F. Siegel. 2001. “The Efficient Use of Conditioning Information in Portfolios.” *Journal of Finance* 56(3): 967–982.
- Filipović, Damir, and Paul Schneider. 2025. “Fundamental Properties of Linear Factor Models.” Swiss Finance Institute Research Paper; SSRN Working Paper 4933856.
- Goyal, Amit, and Ivo Welch. 2008. “A Comprehensive Look at the Empirical Performance of Equity Premium Prediction.” *Review of Financial Studies* 21(4): 1455–1508.

- Goyal, Amit, Ivo Welch, and Athanasse Zafirov. 2024. “A Comprehensive 2022 Look at the Empirical Performance of Equity Premium Prediction.” *Review of Financial Studies* 37(11): 3490–3557.
- Greenwald, Jaffe. 2025. “Anomalies, Roll’s Critique, and Proxy Error.” Working paper, University of Arizona; SSRN Working Paper 4558057.
- Gu, Shihao, Bryan Kelly, and Dacheng Xiu. 2020. “Empirical Asset Pricing via Machine Learning.” *Review of Financial Studies* 33(5): 2223–2273.
- Haddad, Valentin, Serhiy Kozak, and Shrihari Santosh. 2020. “Factor Timing.” *Review of Financial Studies* 33(5): 1980–2018.
- Hansen, Lars Peter, and Scott F. Richard. 1987. “The Role of Conditioning Information in Deducing Testable Restrictions Implied by Dynamic Asset Pricing Models.” *Econometrica* 55(3): 587–613.
- Hasbrouck, Joel. 2009. “Trading Costs and Returns for U.S. Equities: Estimating Effective Costs from Daily Data.” *Journal of Finance* 64(3): 1445–1477.
- He, Zhiguo, Bryan Kelly, and Asaf Manela. 2017. “Intermediary Asset Pricing: New Evidence from Many Asset Classes.” *Journal of Financial Economics* 126(1): 1–35.
- Holm, Sture. 1979. “A Simple Sequentially Rejective Multiple Test Procedure.” *Scandinavian Journal of Statistics* 6(2): 65–70.
- Hong, Harrison, and Jeremy C. Stein. 1999. “A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets.” *Journal of Finance* 54(6): 2143–2184.
- Hong, Harrison, Terence Lim, and Jeremy C. Stein. 2000. “Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies.” *Journal of Finance* 55(1): 265–295.
- Hou, Kewei. 2007. “Industry Information Diffusion and the Lead-Lag Effect in Stock Returns.” *Review of Financial Studies* 20(4): 1113–1138.
- Hou, Kewei, and Tobias J. Moskowitz. 2005. “Market Frictions, Price Delay, and the Cross-Section of Expected Returns.” *Review of Financial Studies* 18(3): 981–1020.
- Hou, Kewei, Chen Xue, and Lu Zhang. 2020. “Replicating Anomalies.” *Review of Financial Studies* 33(5): 2019–2133.
- Jensen, Theis Ingerslev, Bryan Kelly, and Lasse Heje Pedersen. 2023. “Is There a Replication Crisis in Finance?” *Journal of Finance* 78(5): 2465–2518.
- Jensen, Theis Ingerslev, Bryan Kelly, Semyon Malamud, and Lasse Heje Pedersen. 2026. “Machine Learning and the Implementable Efficient Frontier.” *Review of Financial Studies*, forthcoming, advance access. <https://doi.org/10.1093/rfs/hhag022>.

- Kang, Xiaoyu, Bei Chen, and Quan Gan. 2026. “Option-Implied Market Risk Premium and Asset Pricing.” Working paper; SSRN Working Paper 6824883.
- Kelly, Bryan, Semyon Malamud, and Kangying Zhou. 2024. “The Virtue of Complexity in Return Prediction.” *Journal of Finance* 79(1): 459–503.
- Kelly, Bryan T., Seth Pruitt, and Yinan Su. 2019. “Characteristics Are Covariances: A Unified Model of Risk and Return.” *Journal of Financial Economics* 134(3): 501–524.
- Kim, Soohun, Robert A. Korajczyk, and Andreas Neuhierl. 2021. “Arbitrage Portfolios.” *Review of Financial Studies* 34(6): 2813–2856.
- Kozak, Serhiy, Stefan Nagel, and Shrihari Santosh. 2018. “Interpreting Factor Models.” *Journal of Finance* 73(3): 1183–1223.
- Kozak, Serhiy, Stefan Nagel, and Shrihari Santosh. 2020. “Shrinking the Cross-Section.” *Journal of Financial Economics* 135(2): 271–292.
- Lettau, Martin, and Markus Pelger. 2020. “Factors That Fit the Time Series and Cross-Section of Stock Returns.” *Review of Financial Studies* 33(5): 2274–2325.
- Lewellen, Jonathan, Stefan Nagel, and Jay Shanken. 2010. “A Skeptical Appraisal of Asset Pricing Tests.” *Journal of Financial Economics* 96(2): 175–194.
- Lo, Andrew W., and A. Craig MacKinlay. 1990. “When Are Contrarian Profits Due to Stock Market Overreaction?” *Review of Financial Studies* 3(2): 175–205.
- Martin, Ian. 2017. “What Is the Expected Return on the Market?” *Quarterly Journal of Economics* 132(1): 367–433.
- Martin, Ian W. R., and Stefan Nagel. 2022. “Market Efficiency in the Age of Big Data.” *Journal of Financial Economics* 145(1): 154–177.
- McLean, R. David, and Jeffrey Pontiff. 2016. “Does Academic Research Destroy Stock Return Predictability?” *Journal of Finance* 71(1): 5–32.
- Moreira, Alan, and Tyler Muir. 2019. “Should Long-Term Investors Time Volatility?” *Journal of Financial Economics* 131(3): 507–527.
- Muravyev, Dmitriy, Neil D. Pearson, and Joshua M. Pollet. 2025. “Anomalies and Their Short-Sale Costs.” *Journal of Finance* 80(6): 3639–3694.
- Nathan, Daniel, and Matti Suominen. 2026. “The Liquidity-Demand Component of the Factor Zoo.” Working paper; SSRN Working Paper 6909918.

- Novy-Marx, Robert, and Mihail Velikov. 2016. “A Taxonomy of Anomalies and Their Trading Costs.” *Review of Financial Studies* 29(1): 104–147.
- Novy-Marx, Robert, and Mihail Velikov. 2022. “Betting Against Betting Against Beta.” *Journal of Financial Economics* 143(1): 80–106.
- Parsons, Christopher A., Riccardo Sabbatucci, and Sheridan Titman. 2020. “Geographic Lead-Lag Effects.” *Review of Financial Studies* 33(10): 4721–4770.
- Pástor, Ľuboš, and Robert F. Stambaugh. 2003. “Liquidity Risk and Expected Stock Returns.” *Journal of Political Economy* 111(3): 642–685.
- Patton, Andrew J., and Brian M. Weller. 2020. “What You See Is Not What You Get: The Costs of Trading Market Anomalies.” *Journal of Financial Economics* 137(2): 515–549.
- Saxena, Konark, and Mandeep Singh. 2026. “When Tastes Affect Asset Prices But Not Expected Returns.” Working paper; SSRN Working Paper 6167788.
- Smith, Simon C., and Allan Timmermann. 2022. “Have Risk Premia Vanished?” *Journal of Financial Economics* 145(2): 553–576.

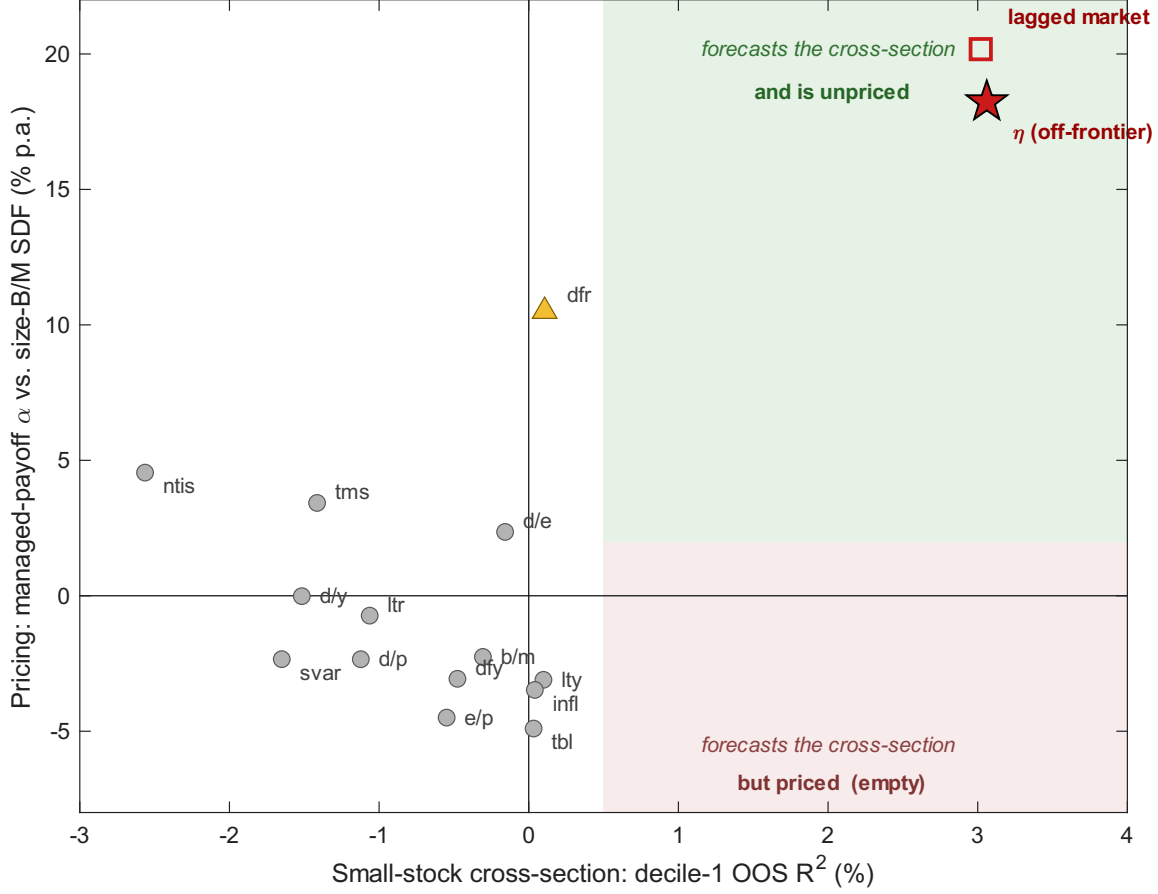


Figure 1: Among standard predictors, only the market's off-frontier component forecasts the small-stock cross-section, and it does so unpriced by the size-value kernel

Notes. Each point is one of the fourteen Goyal and Welch (2008) predictors, plus the off-frontier component η and its proxy the lagged market. The horizontal axis is the recursive out-of-sample R^2 of the smallest size decile's excess return forecast on the lagged predictor (decile-1 OOS R^2); the vertical axis is the annualized alpha of the managed payoff $mp_t = z_{t-1}R_{1,t}^e$ (decile 1 timed on the lagged standardized predictor) against the shrinkage-disciplined 25 size-B/M SDF ($\kappa = 0.01$). Points in the upper-right region forecast the cross-section materially and do so unpriced (large SDF alpha); the lower-right region (forecasts but priced) is empty. Only η and the lagged market forecast the cross-section, both in the unpriced direction; the default return spread (dfr) carries a positive alpha on essentially zero out-of-sample forecast, the persistent-signal artifact of Section 5.1. The shaded regions cover predictors with a decile-1 OOS R^2 above 0.5%, split at an alpha of 2% per year; both thresholds are drawn for display only. A visual restatement of Table 8, Panel B; sample 1963:07–2025:12.

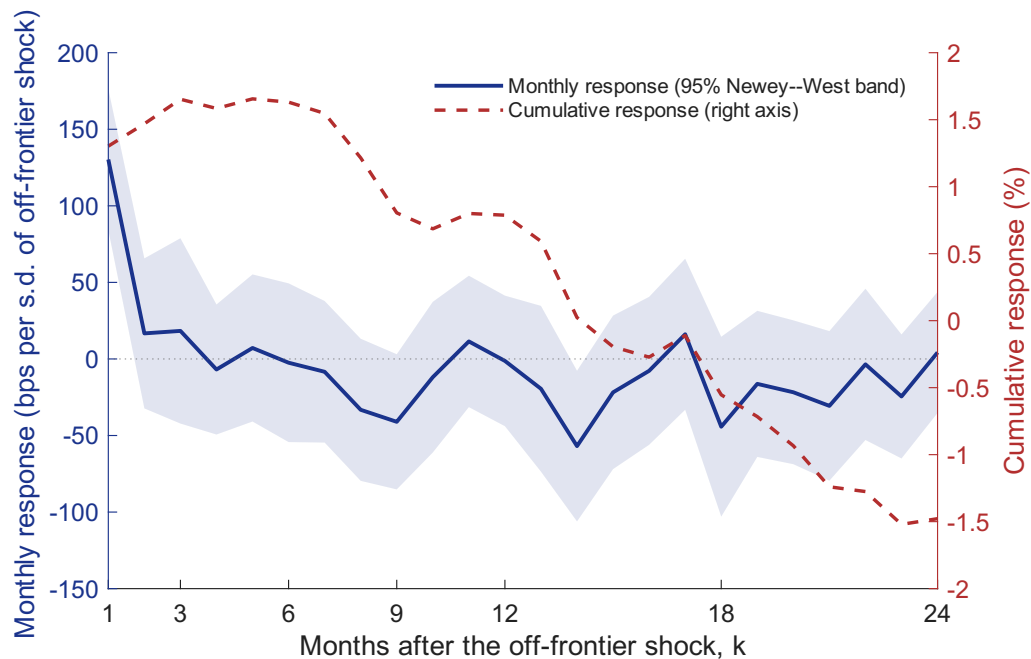


Figure 2: Response of the smallest size decile to an η shock

Notes. The line (left axis) plots the coefficient b_k from regressing the value-weighted decile-1 excess return in month $t+k$ on the standardized off-frontier component η_t , for $k = 1, \dots, 24$; the shaded region is a 95% Newey–West (12-lag) band. The cumulative response $\sum_{j \leq k} b_j$ is on the right axis, in percent. Sample 1963:07–2026:04.

Table 1: Out-of-sample prediction of size deciles by η

size decile	OOS R^2 (%)	Clark–West	in-sample t
1 (small)	3.65	3.9	5.53
2	1.65	2.7	4.14
3	0.37	1.9	3.26
4	−0.07	1.5	2.67
5	−0.09	1.2	2.38
6	−0.21	1.0	2.52
7	−0.08	1.0	2.33
8	−0.03	0.7	1.85
9	−0.15	0.5	1.63
10 (big)	−0.29	−0.4	1.17

Notes. Recursive predictive regressions of each value-weighted size decile’s month- $t+1$ excess return on η_t , with the Hansen–Richard objects and the predictive coefficient estimated through t . OOS R^2 is relative to the prevailing-mean benchmark. Sample 1963:07–2026:04; out-of-sample period begins after a 240-month initial window. The out-of-sample columns re-estimate w , R^{e*} , and η recursively through t ; the in-sample t uses the full-sample η (Newey–West, 12 lags). Clark–West can be positive where the out-of-sample R^2 is negative, because it credits the larger model for forecast content after charging for the estimation noise it adds; this is the expected pattern for the mid deciles (e.g. decile 4, OOS R^2 −0.07%, Clark–West 1.5) and is not a contradiction.

Table 2: The off-frontier component versus the rest of the market

size decile	in-sample t (NW)		OOS R^2 (%)	
	η	$\text{mkt} \perp \eta$	η	$\text{mkt} \perp \eta$
1 (small)	6.11	1.60	3.65	-2.70
2	4.16	0.10	1.65	-1.11
3	3.20	-0.37	0.37	-0.63
4	2.66	-0.10	-0.07	-0.62
5	2.32	-0.44	-0.09	-0.44
6	2.44	-0.68	-0.21	-0.42
7	2.23	-0.97	-0.08	-0.19
8	1.78	-1.02	-0.03	-0.13
9	1.58	-1.31	-0.15	-0.01
10 (big)	1.18	-1.65	-0.29	0.00

Notes. The market excess return is split into η and the frontier component $\text{mkt} \perp \eta = w R^{e*}$, the part orthogonal to η . The in-sample t columns are from the *joint* regression of each decile's month- $t+1$ excess return on both lagged components (Newey–West, 12 lags); for the small deciles η 's joint t exceeds its single-predictor t in Table 1 (6.11 versus 5.53 at decile 1) because the frontier component absorbs residual variance, and the two are nearly identical elsewhere. Each OOS R^2 column is a *separate* single-predictor recursive forecast (the variable named in the column header only), relative to the prevailing mean. Sample 1963:07–2026:04.

Table 3: Priced and unpriced components of η *Panel A: variance and mean decomposition*

	η	priced η_p	unpriced η_u
share of $\text{var}(\eta)$	100%	2.6%	97.4%
mean (% p.a.)	0.00	+3.04	−3.04

Panel B: OOS R^2 (%) predicting size deciles, by component

size decile	η	η_p (priced)	η_u (unpriced)
1 (small)	3.65	−0.95	3.78
2	1.65	−0.63	1.84
3	0.37	−0.36	0.50
10 (big)	−0.29	−0.22	−0.23

Notes. The disciplined SDF is the L^2 -shrinkage tangency of the 25 size-book-to-market portfolios, with $\kappa = 0.01$ fixed ex ante (a split-half validation check selects the same value; Appendix B). The components are $\eta_p = \beta \text{MVE}$ and $\eta_u = \eta - \beta \text{MVE}$. Panel B uses recursive out-of-sample predictive regressions as in Table 1; the decile-1 Clark–West statistic for η_u is 4.0.

Table 4: Alpha of the traded η -timed payoff against each kernel

kernel	priced share	managed-payoff	
	$\text{corr}(\eta, \text{MVE})^2$	α (% p.a.)	NW $t(\alpha)$
CAPM (market)	77.2%	18.5	6.15
FF5 frontier (R^{e*})	0.0%	16.2	5.02
25 size-book-to-market	2.6%	17.3	5.18
disjoint (ind+beta+mom)	8.6%	19.2	5.11
49 industries	15.5%	19.8	5.47
10 beta deciles	37.1%	19.4	5.87
10 momentum deciles	18.3%	17.8	5.28

Notes. The traded payoff is $\text{mp}_t = z_{t-1} R_{1,t}^e$, the smallest size decile timed on the lagged standardized off-frontier component. Its α is the annualized intercept ($1200\times$) from regressing mp on the kernel's unit-variance tangency (MVE) return, Newey-West (12 lags). The “priced share” column repeats the predictor-space figure of Table 5 for comparison; the two need not agree, and the contrast at the CAPM (high predictor-space share, highest payoff alpha) is the point. Common sample 1963:07–2026:04 (intersection with the beta- and momentum-sorted portfolios).

Table 5: Priced share of η across stochastic discount factors

SDF (kernel)	priced share $\text{var}(\eta_p)/\text{var}(\eta)$	overlap with R^{e*} $\text{corr}(\text{MVE}, R^{e*})^2$	market exposure $\text{corr}(\text{MVE}, \text{Mkt})^2$
CAPM (market only)	77.2%	22.8%	100.0%
10 beta-decile tangency	37.1%	23.7%	59.0%
10 momentum-decile tangency	18.3%	9.4%	27.3%
49 industry tangency	15.5%	6.8%	22.2%
disjoint tangency (ind+beta+mom)	8.6%	6.3%	14.3%
25 size-book-to-market tangency	2.6%	24.4%	14.3%
CRRA consumption ($\gamma = 2-50$)	$\leq 0.5\%$	—	$\approx 0\%$
FF5 frontier (R^{e*})	0.0%	100.0%	22.8%

Notes. Priced share is $\text{var}(\eta_p)/\text{var}(\eta)$, equivalently $\text{corr}(\eta, \text{MVE})^2$, the fraction of η 's variance spanned by the SDF return. “Overlap with R^{e*} ” is $\text{corr}(\text{MVE}, R^{e*})^2$, the share of the kernel aligned with the single direction R^{e*} to which η is orthogonal by construction; “market exposure” is $\text{corr}(\text{MVE}, \text{Mkt})^2$. The disjoint tangency uses 67 portfolios sorted only on industry, market beta, and prior-return momentum (of 69 nominal; two industries lack complete data over the sample), none on size or book-to-market, so it shares no sorting variable with η 's construction inputs. Portfolio SDFs use the L^2 -shrinkage tangency (4); the consumption SDF is $m = (C_t/C_{t-1})^{-\gamma}$ with real nondurables-plus-services growth, and for that row the priced share is the squared correlation of η with the kernel itself (an SDF level, not a portfolio return), so the overlap column does not apply. Within the size-value kernel the priced share is 2.6–3.0% across the shrinkage grid $\kappa \in \{0, 0.01, \dots, 20\}$. The priced share is nearly monotone in the kernel's market exposure, because η is 0.88-correlated with the market. Sample 1963:07–2026:04 (1963:07–2023:12 for the consumption kernel).

Table 6: Priced share of η across disjoint kernels

disjoint kernel	assets	priced share	forced overlap	mkt exposure	partial- r ($\eta^\perp, \text{MVE}^\perp$)	95% CI
industry+beta+momentum (baseline)	67	8.6%	6.3%	14.3%	-0.09	[-0.25, 0.07]
49 industries	47	15.5%	6.8%	22.2%	-0.05	[-0.24, 0.12]
10 beta deciles	10	37.1%	23.7%	59.0%	-0.21	[-0.43, 0.08]
10 momentum deciles	10	18.3%	9.4%	27.3%	-0.08	[-0.28, 0.11]
10 short-term-reversal deciles	10	27.0%	24.3%	47.9%	-0.26	[-0.37, -0.04]
10 long-term-reversal deciles	10	33.7%	31.4%	60.5%	-0.34	[-0.53, -0.01]
10 residual-variance deciles	10	10.7%	22.3%	26.3%	-0.30	[-0.50, -0.06]
reversal + residual variance (3 sets)	30	4.3%	22.1%	16.4%	-0.34	[-0.46, -0.11]
all six disjoint sets	97	3.3%	6.3%	7.8%	-0.14	[-0.26, 0.00]

Notes. Each kernel is the L^2 -shrinkage tangency (4) ($\kappa = 0.01$) of portfolios sorted only on the listed characteristics, none on size or book-to-market, so each is disjoint from η 's construction inputs. Short-term reversal (prior 1–0), long-term reversal (prior 60–13), and residual-variance deciles are from Kenneth French's data library; the industry, beta, and momentum sets are as in Table 5. "Assets" counts portfolios with complete data over the common sample (all-missing columns are dropped). "Priced share" is $\text{corr}(\eta, \text{MVE})^2$; "partial- r " is the market-neutral partial correlation $\text{corr}(\eta^\perp, \text{MVE}^\perp)$ of Proposition 1, η and MVE residualized on the market. By that proposition the mechanical benchmark for the priced share is the market-implied share $\text{corr}(\eta, \text{Mkt})^2 \text{corr}(\text{MVE}, \text{Mkt})^2$ (e.g. 11.0% at 67 assets, 6.0% at 97); the priced share lies below it in every row, so partial- $r \leq 0$ throughout and no kernel spans η 's market-neutral variation. The raw priced share falls as the asset space is enlarged (a decline of 5.3 points from 67 to 97 assets, bootstrap 95% CI [0.8, 6.9], $p < 0.001$) because the enlarged tangency is less market-exposed. The last column reports 95% intervals for partial- r from a stationary block bootstrap (mean block six months, 5,000 replications, re-estimating η , each tangency, and the market residualization on every resample). Four intervals lie strictly below zero, the 97-asset interval reaches it ($p(\rho > 0) = 0.03$), and the upper end of the 95% interval for the largest of the nine partial correlations is 0.15, a market-neutral spanned share of at most 2.2%. The baseline priced share is imprecise (95% CI [1.3, 11.2]%) and its excess over the kernel's overlap with R^{e*} is not distinguishable from zero ($p = 0.51$), which is why the argument rests on partial- r and the enlargement, not the single-kernel level. Sample 1963:07–2026:04.

Table 7: An interpretation horse race: what standard tools conclude without the decomposition

<i>Panel A: can conditional beta $\times$ premium generate the lead-lag slope?</i>				
priced factor	b_1 (t)	μ_1 , %/mo (t)	implied slope, %/mo	share of observed
market	0.049 (1.43)	0.17 (0.94)	0.22	17.1%
kernel MVE	-0.001 (-0.55)	3.93 (1.04)	0.02	1.5%
<i>Panel B: alpha of the lagged-market-managed payoff under scaled-factor models</i>				
specification	base factor: market		base factor: kernel MVE	
	α , %/yr (t)	absorbed	α , %/yr (t)	absorbed
unconditional	19.1 (5.3)	—	19.2 (4.9)	—
+ $z^m \cdot f$ (standard test)	13.4 (7.6)	29.5%	17.7 (5.0)	7.7%
+ $z^{fr} \cdot f$ (priced state)	19.6 (6.6)	-2.9%	17.2 (4.9)	10.3%
+ $z^\eta \cdot f$ (unpriced)	13.1 (5.9)	31.5%	19.2 (5.1)	-0.0%

Notes. Panel A: the observed lead-lag slope is the coefficient of the smallest decile's excess return on the lagged standardized market, 1.31% per month per standard deviation (NW $t = 5.49$). The coefficient b_1 is the conditional-beta coefficient from $R_{1,t+1}^e = a + (b_0 + b_1 z_t^m) f_{t+1} + e_{t+1}$; μ_1 the premium-predictability coefficient from $f_{t+1} = \mu_0 + \mu_1 z_t^m + u_{t+1}$; the implied risk-based slope is $b_1 \mu_0 + b_0 \mu_1$, the linear term of the product $(b_0 + b_1 z)(\mu_0 + \mu_1 z)$; the quadratic term enters the linear projection only through the skewness of z^m and is negligible here (of order $b_1 \mu_1$, two orders below the linear term). The kernel MVE is scaled to unit variance, so its μ_1 is not comparable to the market's in magnitude; the share column is. Panel B: annualized intercepts (NW, 12 lags) from regressing the managed payoff $z_t^m R_{1,t+1}^e$ on the base factor f plus the indicated scaled factor. The instruments z^m , z^{fr} , z^η are the lagged standardized market, its priced frontier component wR^{e*} , and its off-frontier component η ($\text{corr}(z^{fr}, z^\eta) = 0.00$); "absorbed" is one minus the ratio of the row's alpha to the unconditional alpha. Sample 1963:07–2026:04 ($T = 754$).

Table 8: The located-predictability battery across the Goyal–Welch predictors

Panel A: does the predictor forecast the priced or the unpriced component of the market?

predictor	priced: frontier wR^{e*}		unpriced: off-frontier η	
	in-sample t	OOS R^2	in-sample t	OOS R^2
dividend–price	0.13	−0.49%	0.39	−1.07%
dividend yield	0.26	−0.57%	0.37	−1.14%
earnings–price	−0.32	−0.42%	0.23	−0.63%
dividend payout	0.81	−0.07%	0.31	−0.83%
stock variance	0.58	−0.09%	−0.48	−3.21%
book-to-market	0.20	−0.47%	−0.05	−0.64%
net equity issuance	−0.58	−0.53%	−0.86	−0.40%
T-bill rate	−0.95	−0.05%	−1.57	0.13%
long-term yield	0.34	0.06%	−1.83	0.31%
long-term return	2.39	−0.27%	1.81	−0.42%
term spread	2.75	0.26%	0.06	−0.97%
default yield spread	1.70	−0.04%	0.42	−0.66%
default return spread	0.00	−0.46%	1.06	−0.34%
inflation	0.59	−0.18%	−1.01	0.04%

Panel B: does the predictor forecast the size cross-section, and is it priced?

predictor	dec.-1 OOS R^2	payoff mean	α vs SDF	NW $t(\alpha)$
		(% p.a.)	(% p.a.)	
dividend–price	−1.12%	−2.1	−2.3	−0.67
dividend yield	−1.52%	−0.6	−0.0	−0.00
earnings–price	−0.55%	−7.6	−4.5	−1.26
dividend payout	−0.16%	5.8	2.4	0.57
stock variance	−1.65%	−1.7	−2.3	−0.22
book-to-market	−0.31%	−2.2	−2.3	−0.69
net equity issuance	−2.56%	1.9	4.5	1.01
T-bill rate	0.03%	−7.7	−4.9	−1.18
long-term yield	0.10%	−6.1	−3.1	−0.68
long-term return	−1.06%	1.3	−0.7	−0.16
term spread	−1.41%	3.2	3.4	0.77
default yield spread	−0.48%	1.4	−3.1	−0.49
default return spread	0.11%	6.5	10.5	2.71
inflation	0.04%	−6.5	−3.5	−0.51
off-frontier η	3.06%	15.2	18.2	3.85
lagged market	3.02%	13.8	20.2	3.77

Notes. The fourteen Goyal and Welch (2008) predictors (lagged), plus the off-frontier component η and the lagged market for reference. *Panel A:* each predictor forecasts the priced frontier component wR^{e*} and the unpriced off-frontier η of the market return; in-sample t is Newey–West (12 lags), OOS R^2 recursive versus the prevailing mean with the Hansen–Richard decomposition re-estimated through t . *Panel B:* decile-1 OOS R^2 is the recursive forecast of the smallest size decile (as in Table 1); the managed payoff is $mp_t = z_{t-1}R_{1,t}^e$, decile 1 timed on the lagged standardized predictor, and α is its annualized intercept on the 25 size–book-to-market ridge SDF (as in Table 4), Newey–West (12 lags). Panel B means and alphas are computed on the common evaluation window 1983:07–2025:12 (the recursive η burn-in) for every row. None of the fourteen forecasts the cross-section; only η (and its proxy the lagged market) does, in the unpriced direction. The default-return-spread row (10.5% alpha on a 0.11% OOS R^2) is the persistent-signal artifact discussed in Section 5.1. The 3.06% for η here is on the common 1963:07–2025:12 sample; cf. 3.65% on the full sample (Table 1). Panel A’s term-spread entry forecasts the priced frontier component wR^{e*} , a small share of market variance, not the

Table 9: Out-of-sample conditional timing by direction

conditioning on	annual mean	annual Sharpe	$t(\text{mean})$
lagged Mkt–RF	14.8%	0.70	4.56
η (full)	15.1%	0.59	3.84
η_u (unpriced)	15.0%	0.61	3.96
η_p (priced)	−0.5%	−0.10	−0.63

Notes. Each strategy weights the ten size deciles by their recursively predicted return given the conditioning variable, with all inputs estimated through t ; the lagged-market row uses month- t Mkt–RF as the timing variable. The last column is the i.i.d. t on the strategy mean; the text reports the Newey–West t (lagged Mkt–RF: 4.84; η_u : 5.17) and a stationary block-bootstrap 95% interval (lagged Mkt–RF: [0.43, 0.99]; η_u : [0.38, 0.91]). The lagged-market and η_u timing strategies are 0.81 correlated; the priced strategy is 0.32 correlated with the lagged-market strategy. Annual means are on a common weight scale; because each strategy is a scale-arbitrary zero-cost overlay, the defensible object is the out-of-sample Sharpe. The in-sample maximum Sharpe ratio of the conditional frontier (about 1.0 versus 0.53 unconditional) is upward biased and is not the basis for inference. Sample 1963:07–2026:04; out-of-sample from 1983:07 (240-month initial window).

Table 10: Near-arbitrage and the friction wedge

η -timed small-decile alpha vs. market + SDF jointly	19.1%/yr ($t = 4.1$)
gross annual Sharpe of strategy	0.63
breakeven one-way cost	≈ 123 bps
<i>net annual Sharpe by one-way cost:</i>	
25 bps	0.50
50 bps	0.37
100 bps	0.12
200 bps	-0.39

Notes. The traded strategy times the smallest size decile on real-time η ; net returns subtract one-way cost $\times$ turnover ($|z_t - z_{t-1}|$). This is the single tradeable object I cost in the ladder; the cross-sectional overlays of Table 9 and the managed decile payoffs hold the same microcap names and face costs that bind sooner (the all-decile overlay’s breakeven one-way cost is 35 basis points, the ex-microcap version’s 22, against this strategy’s 123). The cost calculation omits reconstitution turnover, long-short financing and borrow on the (largely untradeable) short leg, and price impact, so the net Sharpe ratios overstate what is attainable; this strengthens, not weakens, the “unexploitable” reading. The alpha row regresses the real-time timed payoff on the market and the disciplined SDF jointly (Newey–West, 12 lags). Sample 1963:07–2026:04; out-of-sample from 1983:07.

Appendix A. The Hansen–Richard estimator

Gross construction. Stack the K base returns in R ($T \times K$). The minimum-second-moment payoff that prices them is $x^* = R(R^\top R/T)^{-1}\mathbf{1}_K$ with price $p(x^*) = \frac{1}{T}x^{*\top}x^*$, and $R^* = x^*/p(x^*)$. Excess returns are $R^e = R - R^*\mathbf{1}_K^\top$ (drop one linearly dependent column), and $R^{e*} = R^e(R^{e\top}R^e)^{-1}R^{e\top}\mathbf{1}_T$ is the fitted value of a regression of $\mathbf{1}_T$ on R^e with no intercept. For asset i , $w_i = (\bar{R}_i - \bar{R}^*)/\bar{R}^{e*}$ and $n_i = R_i - R^* - w_i R^{e*}$.

Market excess return. Because Mkt–RF has price zero its R^* loading is zero, and (1) reduces to Mkt–RF = $w R^{e*} + \eta$ with R^{e*} as in (2). The estimator (3) uses this two-term form with the factor excess returns as R^e . Because the market is an excess return, η is computed entirely in excess-return space from (2)–(3). The gross R^* objects above are shown only to place η within the full decomposition and are not used to form it (in particular, the zero-cost factors are not converted to gross returns). All reported results use this excess construction, for which $\text{corr}(\eta, \text{Mkt}) = 0.88$.

Properties. R^{e*} represents means: $E[R^{e*}R_j^e] = E[R_j^e]$ for any excess return R_j^e , and $E[(R^{e*})^2] = E[R^{e*}]$. Hence $E[\eta R^{e*}] = E[\text{Mkt} - \text{RF}] - w E[R^{e*}] = 0$, and with $\bar{\eta} = 0$, $\text{cov}(\eta, R^{e*}) = 0$, which gives the variance decomposition in Section 2.

Recursive estimator. For the out-of-sample tests, at each month $t \geq t_0 = 240$ the objects R^{e*} , w , η are recomputed from (2)–(3) on $R_{1:t}^e$. The value η_t uses only information through t and is the predictor for $t+1$. The linear weight vector c with $\eta = R^e c$, used to form daily η in the Internet Appendix, is recovered by ordinary least squares of η on R^e within the window (exact, since η lies in the span of R^e).

The location decomposition. The decomposition of Section 2.3, stated formally. The split (5) is a centered projection, so the orthogonality statements below use the covariance inner product, while the Hansen–Richard objects of (1) live in the uncentered metric. Write $m \equiv \text{MVE}$, $\eta = \eta_p + \eta_u$ with $\eta_p = \beta m$, $\beta = \text{Cov}(\eta, m)/\text{Var}(m)$, and $g_{i,k} = \text{Cov}(\eta_{k,t}, R_{i,t+1}^e)$ for $k \in \{p, u\}$.

Proposition 2. (i) $\text{Cov}(\eta_{p,t}, \eta_{u,t}) = 0$, so the priced and unpriced components are uncorrelated predictors. In the joint regression of any decile on both, the multiple-regression slopes equal the univariate slopes and the predictive R^2 splits additively, $R_i^2(\eta_p, \eta_u) = R_i^2(\eta_p) + R_i^2(\eta_u)$, with $g_i = g_{i,p} + g_{i,u}$ by linearity. (ii) $\eta_p \in \text{span}\{m\}$, so the expected-return variation it indexes is a time-varying loading on the priced factor. The unpriced part η_u has zero beta on m , so the variation it indexes improves the conditional frontier through a direction m does not price. (iii) The first-order (mean-timing) conditioning gain is priced to the extent $g_{i,p} \neq 0$ and unpriced to the extent $g_{i,u} \neq 0$. If $g_{i,p} = 0$ for all i while $g_i \neq 0$ for some i , the mean of every improving mean-timing payoff comes entirely from the unpriced direction.

Proof. (i) $\beta = \text{Cov}(\eta, m)/\text{Var}(m)$ gives $\text{Cov}(\eta_u, m) = \text{Cov}(\eta, m) - \beta \text{Var}(m) = 0$ and $\text{Cov}(\eta_p, \eta_u) = \beta[\text{Cov}(m, \eta) - \beta \text{Var}(m)] = 0$. Uncorrelated regressors give multiple-regression slopes equal to the univariate slopes and additive explained variance, and $g_i = g_{i,p} + g_{i,u}$ follows from $\eta = \eta_p + \eta_u$. (ii) $\eta_p = \beta m \in \text{span}\{m\}$ by construction, and the premium m assigns a return is governed by its beta on m , zero for η_u by (i). (iii) Timing decile i by the demeaned signal $\propto \eta_{k,t}$ gives a payoff with unconditional mean $g_{i,k}$, so if $g_{i,p} = 0$ for all i the mean-timing gain comes entirely from η_u . Whether the timing *payoff* $\eta_{u,t} R_{i,t+1}^e$ is itself orthogonal to m is a conditional-second-moment question the proposition does not address. It holds if $\text{Cov}_t(R_{i,t+1}^e, m_{t+1})$ does not move with $\eta_{u,t}$ and $\eta_{u,t}$ does not forecast m_{t+1} , conditions the data can violate, which is why Section 5.1 measures the traded payoff's kernel loadings empirically rather than asserting them. $\square$

Appendix B. Estimation details

The w estimator. The scalar $w = \overline{\text{Mkt}} - \overline{\text{RF}}/\overline{R^{e*}}$ is a ratio of two estimated means, and R^{e*} is an in-sample projection, so a natural worry is that η 's early-sample path reflects estimation error rather than the market's frontier deviation. It does not. The real-time η (recursive w and R^{e*} , re-estimated through t) correlates 0.978 with the full-sample η and 0.941 with a version that holds w at its full-sample value. The correlation is 0.975 in the earliest third of the sample, no worse than the 0.992 in the latest third, so the estimator is not noisier where it matters. The first-decile out-of-sample R^2 is 3.65% under recursive w and 3.25% holding w fixed. The series η is insensitive to how w is handled.

SDF shrinkage. The SDF weights solve (4). To avoid any look-ahead, I fix $\kappa = 0.01$ *ex ante* for all out-of-sample statistics. This conventional value is not tuned on the data used for those statistics. As a check, a split-half validation (split the full sample in half, estimate b on each half, evaluate the tangency Sharpe on the other, over the grid $\{0, 0.01, 0.05, 0.2, 1, 5, 20\}$) selects the same $\kappa = 0.01$ (in-sample annual Sharpe 1.38, held-out 0.83; the raw $\kappa = 0$ tangency overfits, held-out 0.81). The split serves only to check the ex-ante choice. The SDF itself is then re-estimated on the full sample at $\kappa = 0.01$. The size-value priced/unpriced split (5) is essentially invariant across the grid (priced share 2.6–3.0%), and κ enters neither η nor the size-decile forecasts (only the split and the conditional timing), so the no-later-data property of the out-of-sample results is secure. The SDF scaling (unit variance), the standardization of conditioning variables, the recursive split into η_p and η_u , the timing weights, and the daily coefficient vector c_t used for realized *dis* are all computed from data through t only.

Newey–West. Predictive-regression t -statistics use the heteroskedasticity- and autocorrelation-

consistent estimator with a Bartlett kernel and 12 lags, $\hat{V} = (Z^\top Z)^{-1} \hat{S} (Z^\top Z)^{-1}$ with $\hat{S} = \sum_{t=1}^T z_t z_t^\top u_t^2 + \sum_{l=1}^{12} (1 - \frac{l}{13}) \sum_{t=l+1}^T (z_t z_{t-l}^\top + z_{t-l} z_t^\top) u_t u_{t-l}$.

Out-of-sample R^2 and Clark–West. For forecast $\hat{y}$ and benchmark $\bar{y}$ (the prevailing mean), $R_{oos}^2 = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2$. The Clark–West statistic for a model that nests the benchmark is the t -statistic on the mean of $f_t = (y_t - \bar{y}_t)^2 - [(y_t - \hat{y}_t)^2 - (\bar{y}_t - \hat{y}_t)^2]$, referred to a one-sided normal.

Conditional timing (Table 9). For a conditioning variable z (standardized through t), at each t I regress each decile’s month- $s+1$ excess return on z_s for $s \leq t-1$, form predicted returns $\hat{r}_i = \hat{\beta}_i z_t$, set cross-sectionally demeaned weights $\omega_i = \hat{r}_i - \bar{\hat{r}}$, and record the strategy return $\sum_i \omega_i R_{i,t+1}$; Sharpe ratios are annualized by $\sqrt{12}$.

η -timing and costs (Table 10). The managed portfolios are $z_{t-1} R_{i,t}^e$ with z the standardized η . The traded strategy times the smallest decile ($z_{t-1} R_{1,t}^e$); monthly turnover is $|z_t - z_{t-1}|$, the net return is gross $- c \times \text{turnover}$ for one-way cost c , and the breakeven cost is mean gross return over mean turnover.

Sharpe-ratio map. Table 11 collects every Sharpe ratio quoted in the paper with its construction, information set, and sample.

Table 11: Map of the Sharpe ratios quoted in the paper

Sharpe	object	construction and information set	reported
<i>Conditional overlays on the ten size deciles (recursive, out of sample 1983:07–2026:04)</i>			
0.61	timing on η_u (headline)	demeaned predicted-return weights, all inputs through t	Table 9
0.59	timing on full η	same rule	Table 9
−0.10	timing on η_p	same rule	Table 9
0.70	timing on lagged Mkt−RF	same rule; 0.81 correlated with the η_u strategy	Table 9
0.43	timing on η , ex-microcap	same rule on deciles 2–10 only	Section 6.2
<i>Ferson–Siegel optimal conditional weights (recursive, same window)</i>			
0.70	FS weights on full η	$w \propto \Sigma^{-1}\mu(z_t)/(1 + \mu^\top \Sigma^{-1}\mu)$, constant κ -shrunk Σ	Section 6.1
0.67, 0.30	FS weights on η_u , on η_p	same rule by conditioning direction	Section 6.1
0.62	naive linear rule on η	$w \propto \Sigma^{-1}\mu(z_t)$	Internet Appendix
<i>Static size-decile frontiers</i>			
0.53	unconditional tangency, in sample	full-sample weights, 1963:07–2026:04	Table 9 notes
0.56	oracle static, out of sample	full-sample look-ahead weights held over the out-of-sample window	Section 6.1
0.16	implementable static, out of sample	recursive κ -shrunk tangency	Section 6.1
≈ 1.0	conditional frontier, in sample	upward-biased maximum, not used for inference	Table 9 notes
<i>Single-decile timed strategy, the costed object (out of sample 1983:07–2026:04)</i>			
0.63	gross, recursive z	$z_{t-1}R_{1,t}^e$ with η standardized through t	Table 10
0.64	gross, full-sample z	in-sample standardization, the managed-payoff convention	Section 5.1
0.50 to −0.39	net of costs	one-way cost (25–200 bps) $\times$ turnover	Table 10
0.75, 0.53	gross by era	pre- and post-decimalization subperiods	Section 8.2
<i>Pricing kernel (1963:07–2026:04)</i>			
1.38, 0.83	disciplined SDF tangency	25 size-B/M portfolios, $\kappa = 0.01$; in sample and split-half held out	Appendix B

Notes. All Sharpe ratios are annualized. The overlays are zero-cost and scale-arbitrary, so their Sharpe ratios rather than their means are the comparable objects. “Recursive” means every input (including η , the SDF, the priced/unpriced split, and any standardization) is estimated from data through t only.